

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7280

Nombre de lignes df clean	7280
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5096
Taille dataset test	2184
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_24h_mean, r_NH3_st_24h_std, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_max, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_24h_mean, r_temperature_st_24h_std, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_temperature_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_24h_mean, r_umidity_st_24h_std, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_min, r_umidity_st_24h_max, r_umidity_st_24h_diff_last, r_umidity_st_24h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

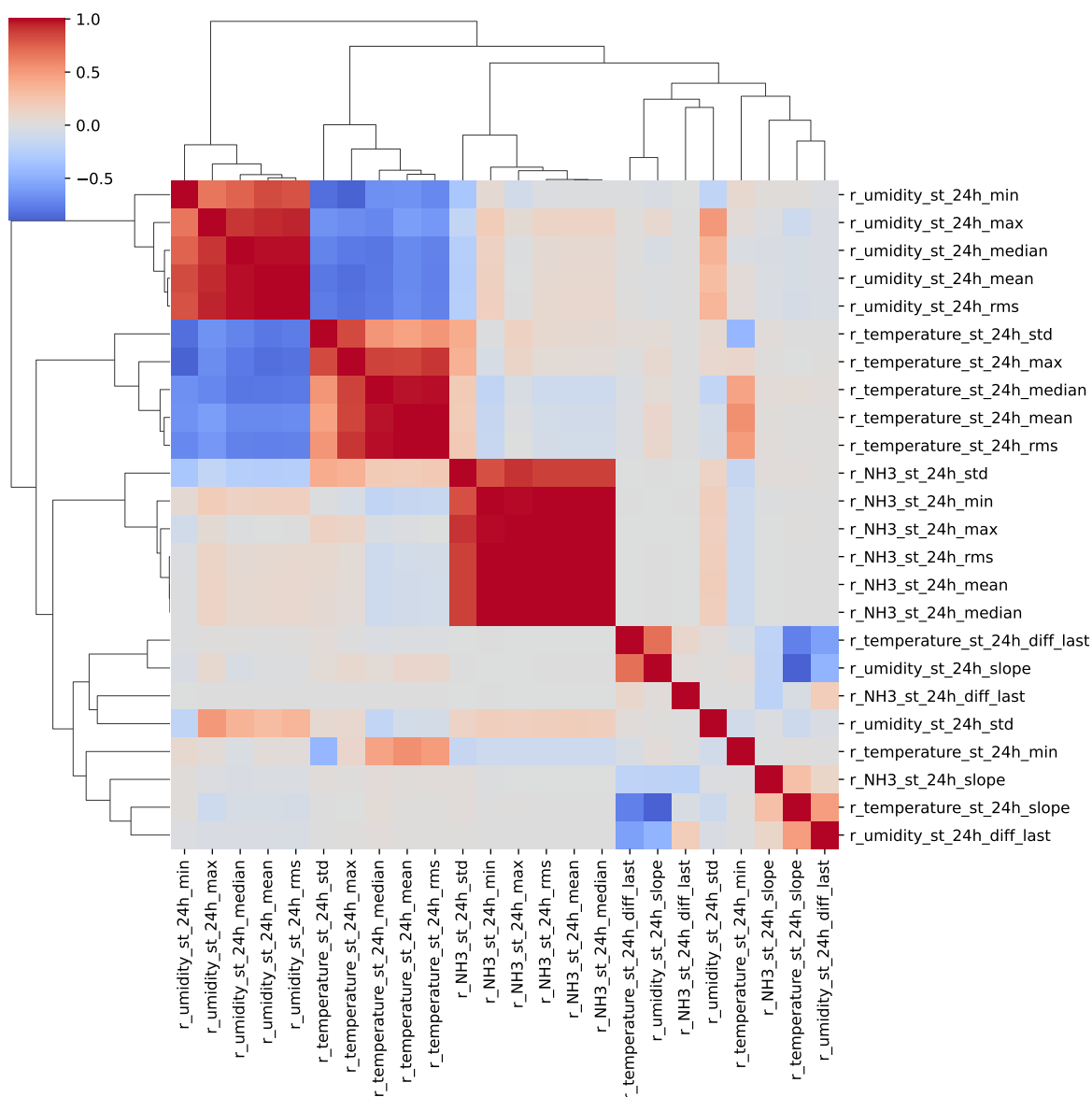
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_24h_mean, r_NH3_st_24h_max, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_std ; **cluster 7:** r_NH3_st_24h_diff_last ; **cluster 8:** r_NH3_st_24h_slope ; **cluster 9:** r_umidity_st_24h_mean, r_temperature_st_24h_mean, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_max, r_temperature_st_24h_std, r_temperature_st_24h_median, r_umidity_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_rms ; **cluster 10:** r_temperature_st_24h_min

; **cluster 11:** r_temperature_st_24h_diff_last ; **cluster 12:** r_umidity_st_24h_slope, r_temperature_st_24h_slope ; **cluster 13:** r_umidity_st_24h_std ; **cluster 14:** r_umidity_st_24h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_24h_mean, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_umidity_st_24h_mean, r_temperature_st_24h_min, r_temperature_st_24h_diff_last, r_umidity_st_24h_slope, r_umidity_st_24h_std, r_umidity_st_24h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_24h_mean, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_umidity_st_24h_mean, r_tempera-

ture_st_24h_min, r_temperature_st_24h_diff_last, r_umidity_st_24h_slope, r_umidity_st_24h_std, r_umidity_st_24h_diff_last

Len(Features_X): 14

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

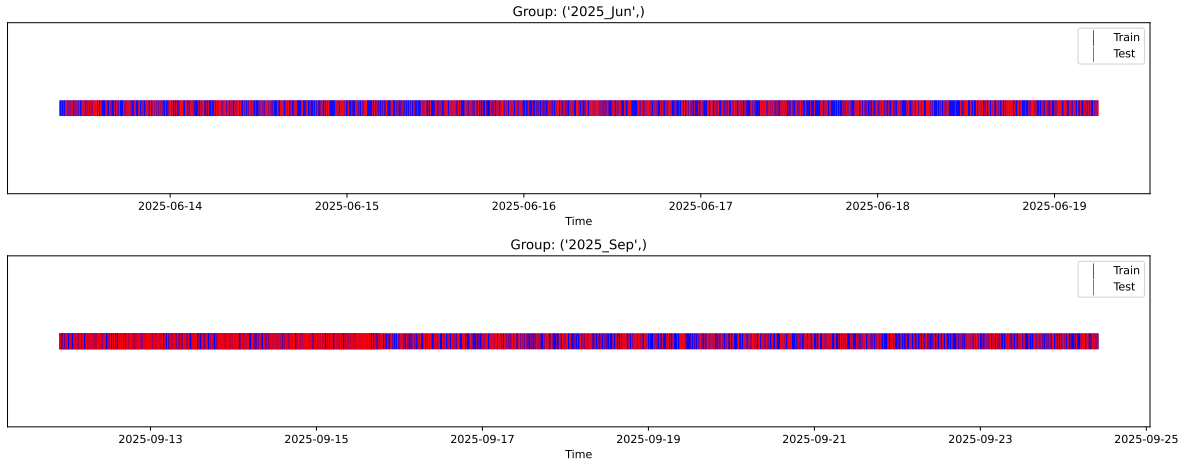
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

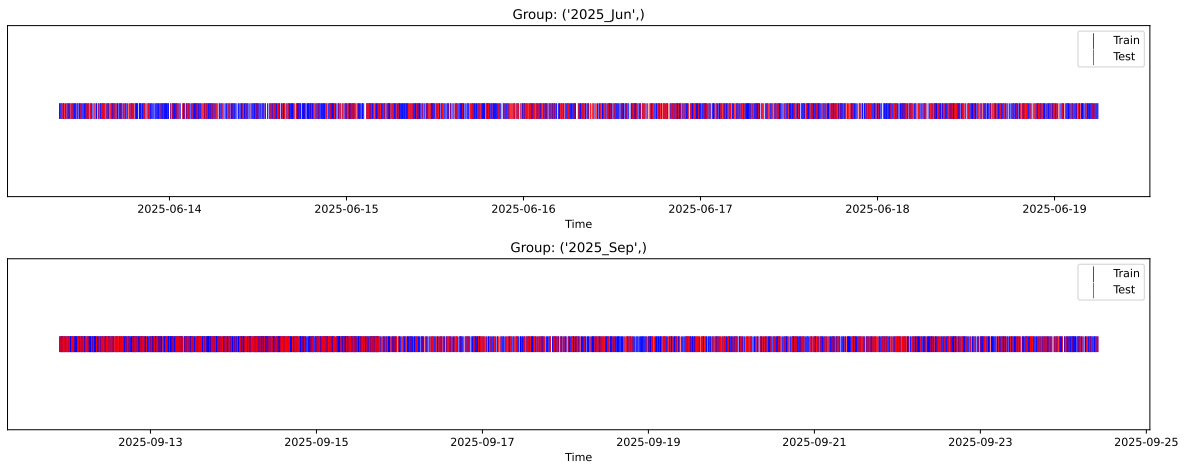
External Split: Train: 5096 samples Test: 2184 samples



3.3.2 Internal CV Folds

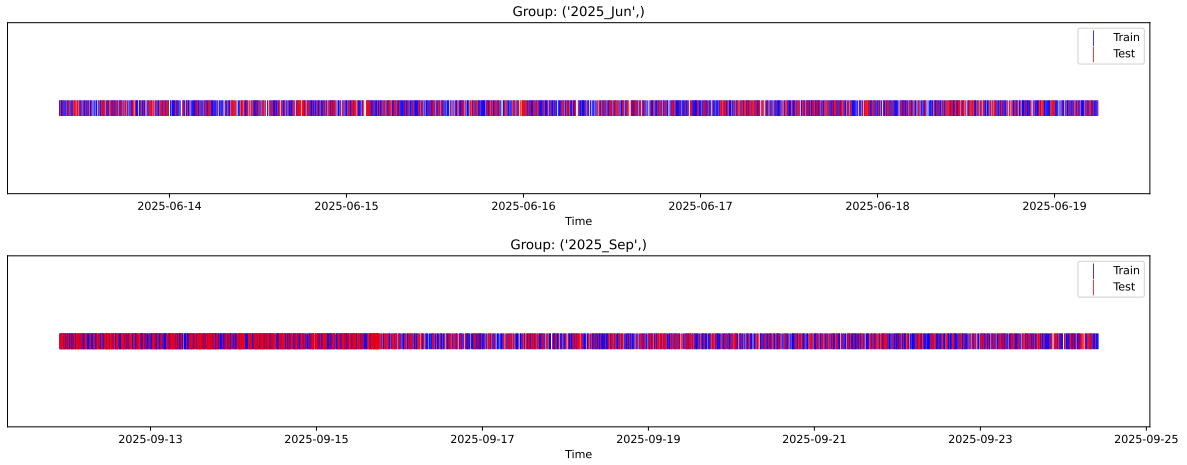
Fold 1/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



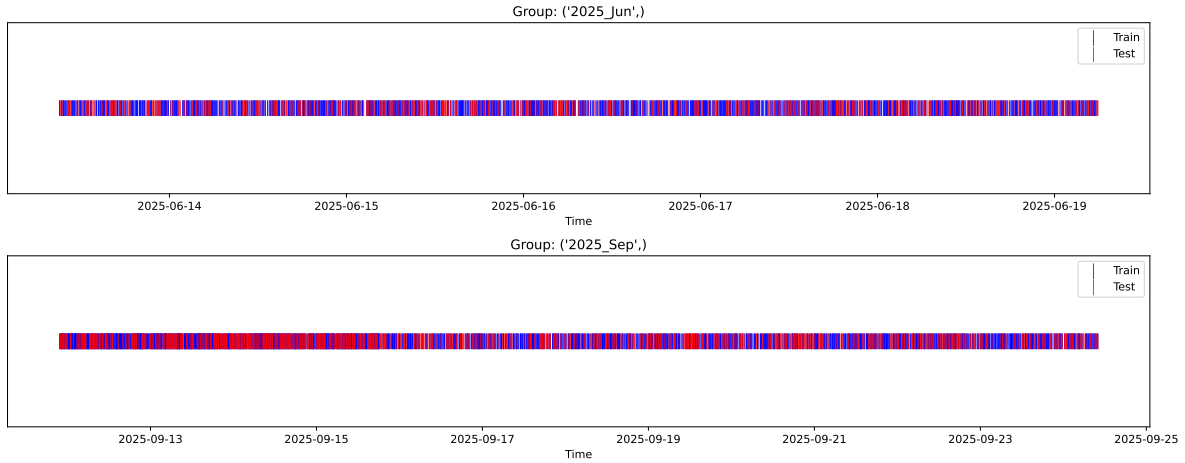
Fold 3/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



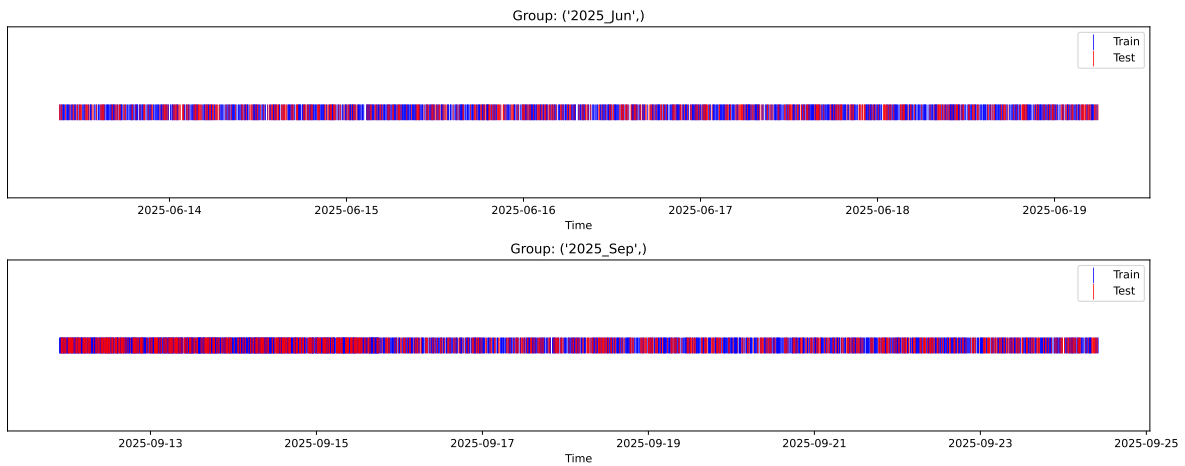
Fold 4/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.362, std=0.816, min=0.959, max=13.845
 y_{test} : mean=2.359, std=0.781, min=1.062, max=7.822

Fold 0 — y_{test} : mean=2.36, std=0.87, y_{train} : mean=2.36, std=0.79

Fold 1 — y_{test} : mean=2.38, std=0.84, y_{train} : mean=2.36, std=0.81

Fold 2 — y_{test} : mean=2.36, std=0.83, y_{train} : mean=2.36, std=0.81

Fold 3 — y_test: mean=2.36, std=0.81, y_train: mean=2.36, std=0.82

Fold 4 — y_test: mean=2.39, std=0.85, y_train: mean=2.35, std=0.80

train scores CV: [0.54068786 0.53618891 0.5249865 0.50059261 0.53299927]

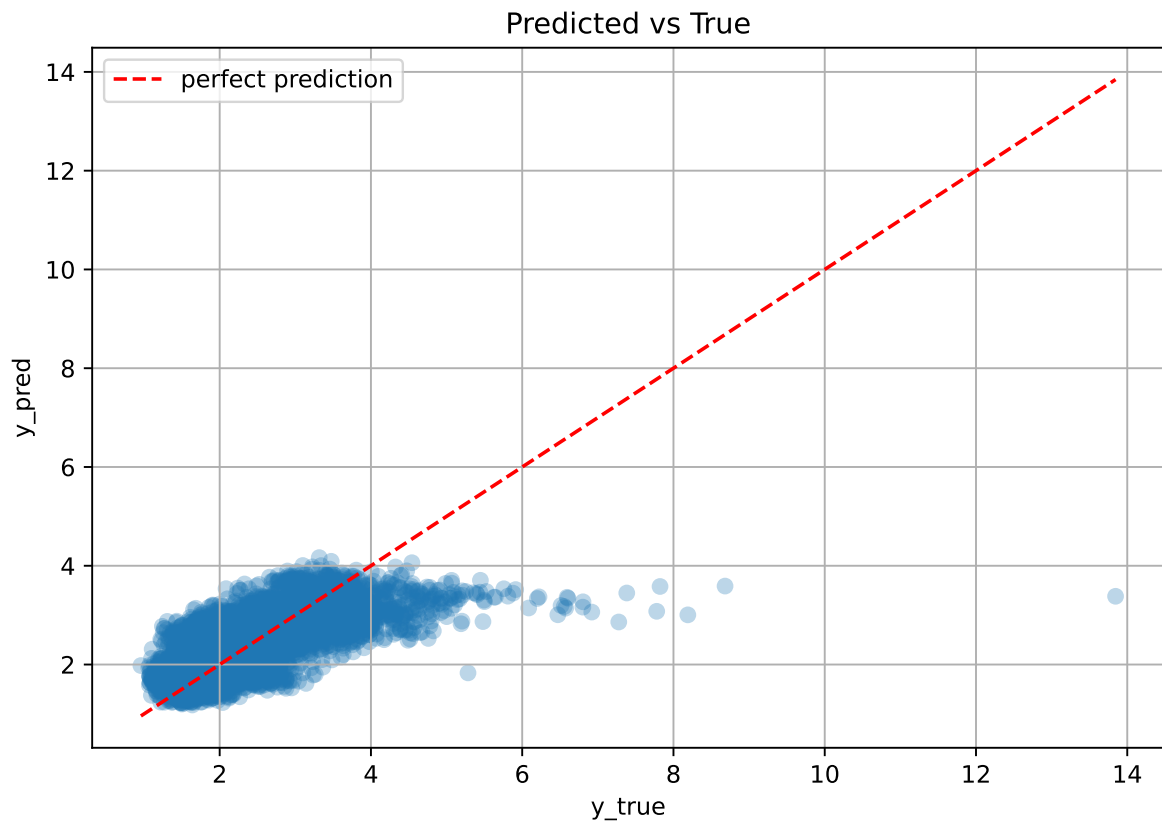
test scores CV: [0.45561122 0.46046057 0.48298483 0.5394142 0.46806407]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5271	0.4813	0.0305	0.5427	0.5432	0.0619	1.1287	1.0951

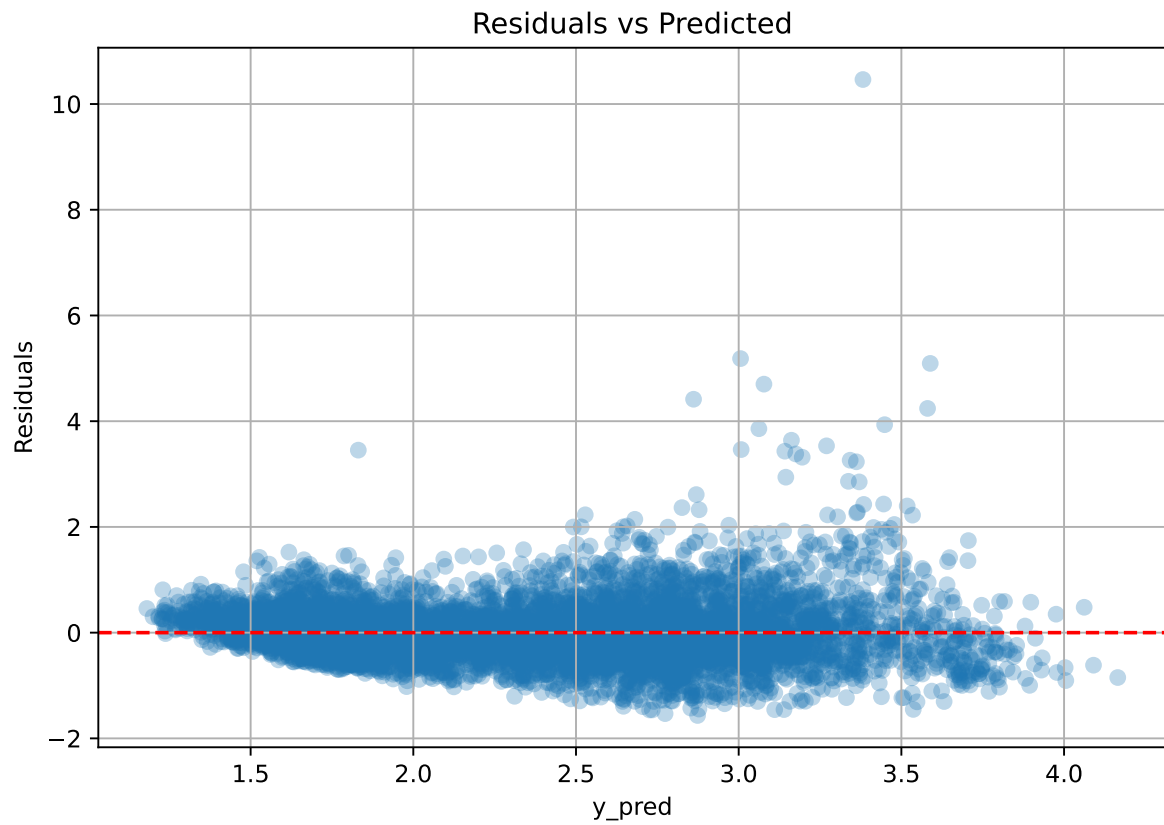
Coefficients:

coef_r_CO2 = -0.0614, coef_r_NH3 = 0.2246, coef_r_NH3_st_24h_diff_last = 0.0216, coef_r_NH3_st_24h_mean = -0.2711, coef_r_NH3_st_24h_slope = -0.0048, coef_r_THI = 1.3265, coef_r_temperature = -1.5872, coef_r_temperature_st_24h_diff_last = 0.1314, coef_r_temperature_st_24h_min = -0.0503, coef_r_umidity = -0.7910, coef_r_umidity_st_24h_diff_last = -0.0875, coef_r_umidity_st_24h_mean = -0.0516, coef_r_umidity_st_24h_slope = 0.1672, coef_r_umidity_st_24h_std = -0.0046, intercept = 2.3622,

3.4.2 Scatter



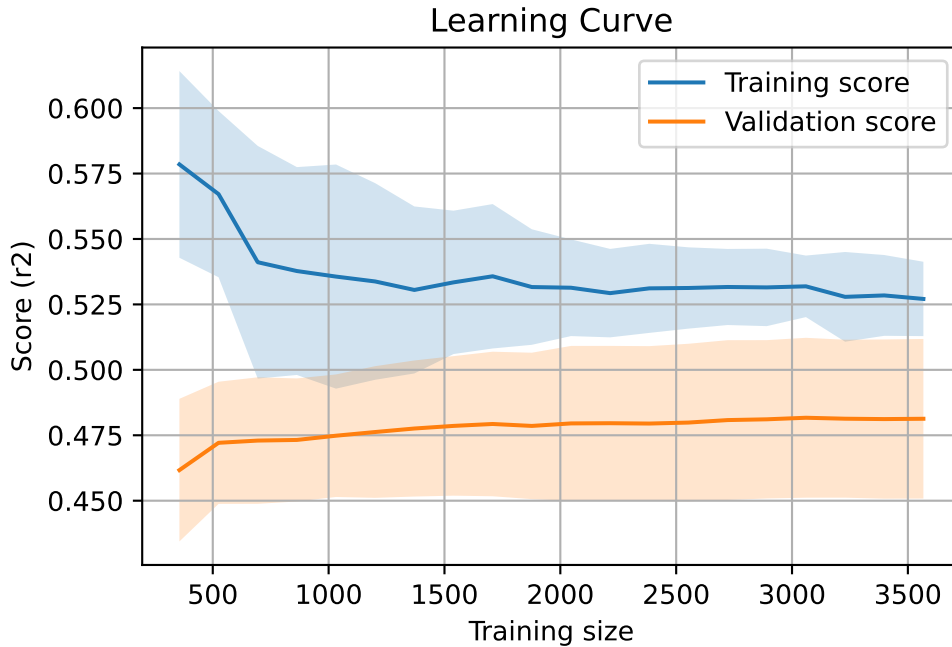
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7280, `Len(training)` : 5096,

`Len(training_real)` : 3568, `Len(test_of_training)` : 1528, `Len(test)` : 2184

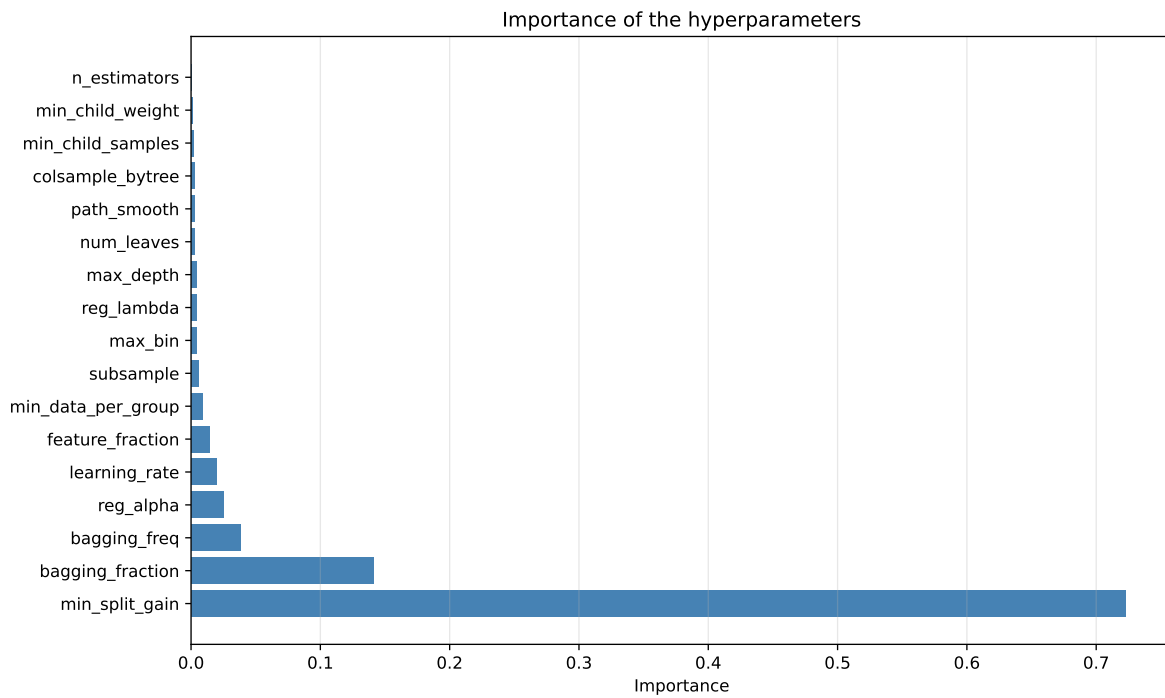
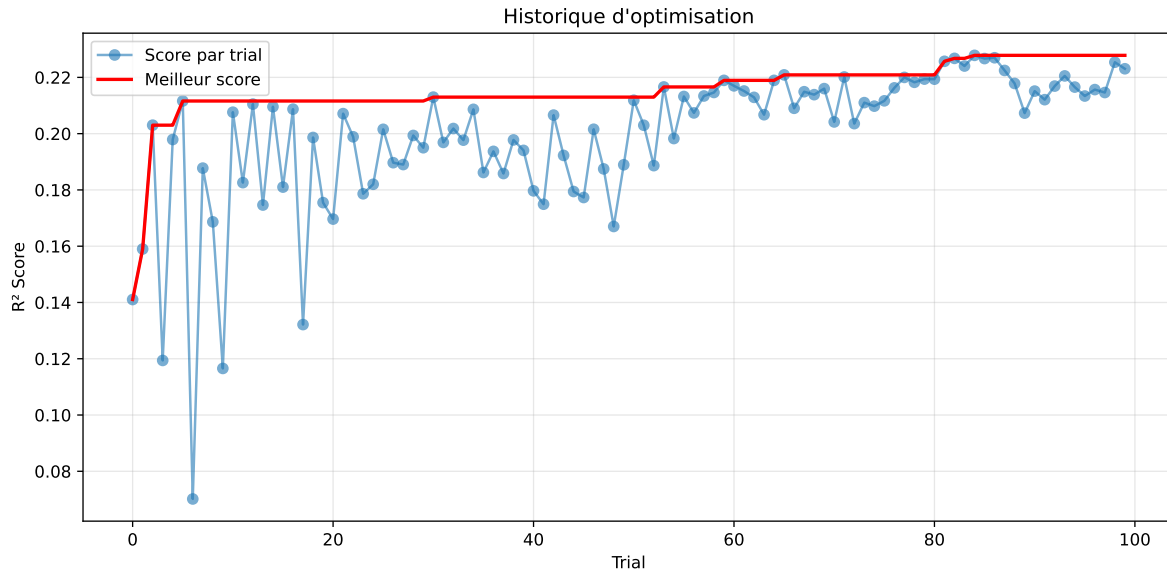


3.5 Model : LGBM

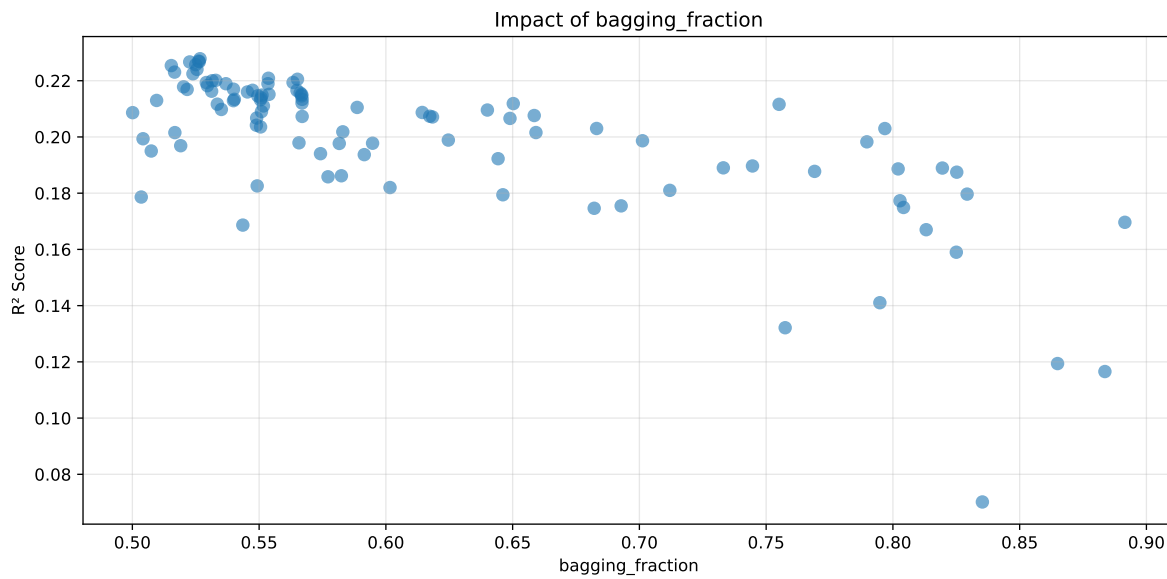
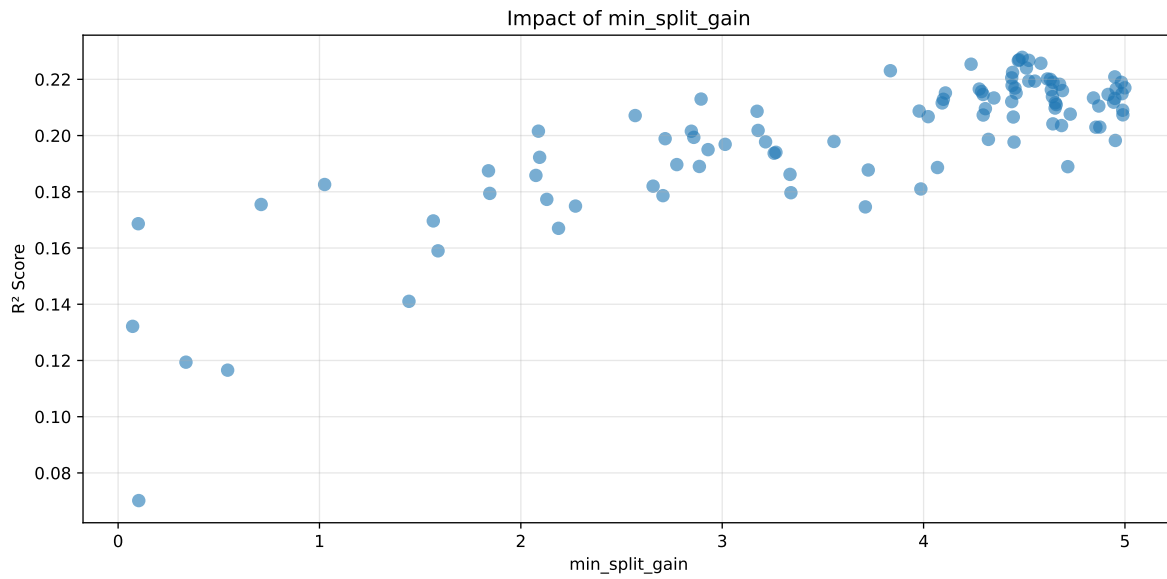
3.5.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.362, std=0.816, min=0.959, max=13.845
y_test: mean=2.359, std=0.781, min=1.062, max=7.822

===== TOP Importance FEATURES =====
feature importance r_temperature_st_24h_diff_last 72 r_umidity 69 r_tempera-
ture_st_24h_min 57 r_temperature 56 r_NH3_st_24h_mean 49 r_CO2 45 r_THI 42 r_umid-
ity_st_24h_mean 40 r_umidity_st_24h_diff_last 32 r_NH3 29 r_umidity_st_24h_slope 26
r_umidity_st_24h_std 25 r_NH3_st_24h_diff_last 16 r_NH3_st_24h_slope 9

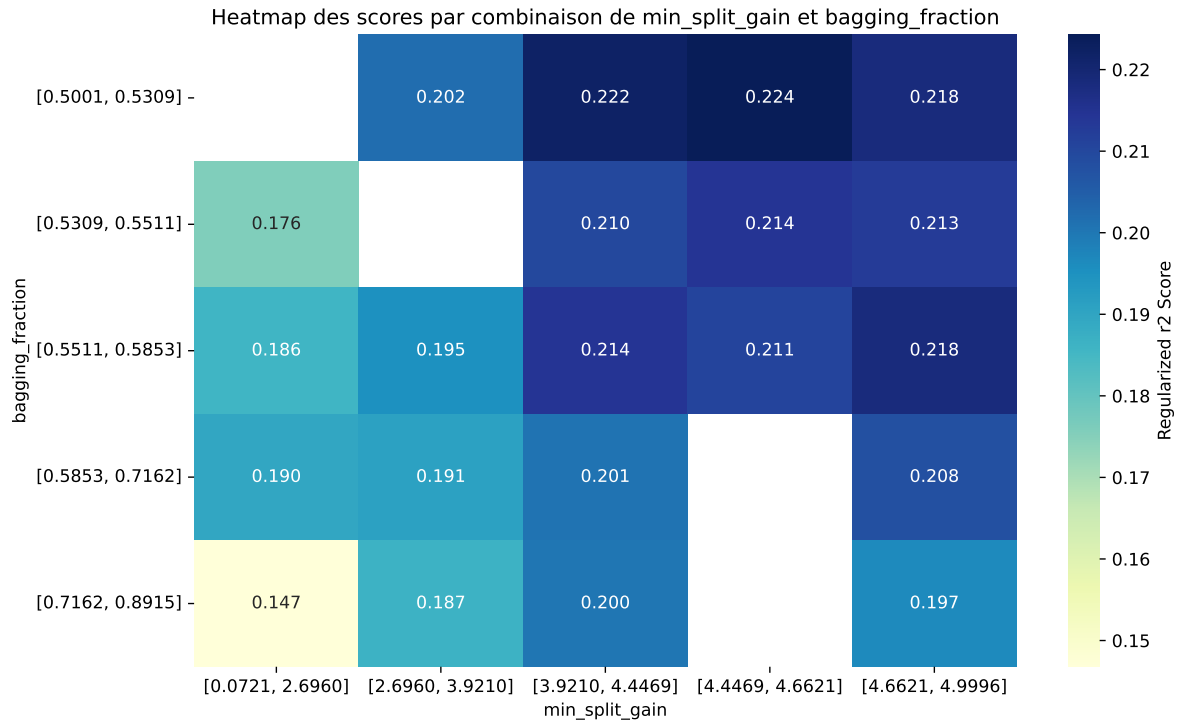


Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6046	0.5418	0.616	0.0298	0.2278	0.6394	0.0976	1.1801	1.1159
2	0.6101	0.5462	0.6205	0.0286	0.227	0.6358	0.0895	1.1639	1.1169
3	0.611	0.5469	0.6212	0.0308	0.2268	0.6332	0.0863	1.1578	1.1171
4	0.6093	0.5455	0.6198	0.0309	0.2267	0.6411	0.0956	1.1753	1.1169
5	0.6047	0.5415	0.6171	0.0292	0.2257	0.6403	0.0988	1.1824	1.1166
6	0.6121	0.5476	0.6226	0.0288	0.2254	0.6473	0.0996	1.182	1.1177
7	0.6094	0.5452	0.6197	0.0251	0.224	0.6349	0.0897	1.1645	1.1178
8	0.6105	0.5459	0.6204	0.0279	0.2231	0.6482	0.1023	1.1874	1.1183
9	0.6138	0.5486	0.6235	0.0322	0.2225	0.6514	0.1028	1.1874	1.1189
10	0.6109	0.5459	0.6216	0.0299	0.2209	0.6475	0.1016	1.1861	1.1191
---	---	---	---	---	---	---	---	---	---
100	0.8301	0.7034	0.7792	0.0315	0.0702	0.8012	0.0977	1.1389	1.1801

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.03143314465050743, 'reg__num_leaves': 46, 'reg__subsample': 0.6507331057776738, 'reg__colsample_bytree': 0.5763165027979588, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.943001543479525, 'reg__reg_lambda': 23.7736258116374, 'reg__min_split_gain': 4.489766386260917, 'reg__path_smooth': 2.574736707708422, 'reg__min_child_weight':

3.318828391698546, 'reg__feature_fraction': 0.5961064862802907, 'reg__bagging_fraction': 0.5266924489190247, 'reg__bagging_freq': 2, 'reg__max_bin': 79, 'reg__min_data_per_group': 192}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.032430487460135576, 'reg__num_leaves': 47, 'reg__subsample': 0.527181543816507, 'reg__colsample_bytree': 0.5904469735440128, 'reg__min_child_samples': 12, 'reg__reg_alpha': 8.966991720379356, 'reg__reg_lambda': 22.18164091513432, 'reg__min_split_gain': 4.47443540480659, 'reg__path_smooth': 3.237983052002873, 'reg__min_child_weight': 3.514015402358447, 'reg__feature_fraction': 0.6073801107754654, 'reg__bagging_fraction': 0.5261344138665431, 'reg__bagging_freq': 2, 'reg__max_bin': 79, 'reg__min_data_per_group': 191}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.031857685254509965, 'reg__num_leaves': 46, 'reg__subsample': 0.6510011969074592, 'reg__colsample_bytree': 0.565830547955404, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.980534345261269, 'reg__reg_lambda': 24.04104200929331, 'reg__min_split_gain': 4.522608716322509, 'reg__path_smooth': 2.605793677539502, 'reg__min_child_weight': 2.9078743505124325, 'reg__feature_fraction': 0.6168357742345726, 'reg__bagging_fraction': 0.5264044323609719, 'reg__bagging_freq': 2, 'reg__max_bin': 80, 'reg__min_data_per_group': 190}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.03401000424427491, 'reg__num_leaves': 46, 'reg__subsample': 0.5314150568468383, 'reg__colsample_bytree': 0.5708888181367114, 'reg__min_child_samples': 13, 'reg__reg_alpha': 8.984777868211832, 'reg__reg_lambda': 23.911940504925653, 'reg__min_split_gain': 4.470384256699584, 'reg__path_smooth': 3.269445277766061, 'reg__min_child_weight': 2.973351562696488, 'reg__feature_fraction': 0.6051443504047754, 'reg__bagging_fraction': 0.522612189743627, 'reg__bagging_freq': 2, 'reg__max_bin': 78, 'reg__min_data_per_group': 191}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.024040425832539545, 'reg__num_leaves': 46, 'reg__subsample': 0.6484211169397059, 'reg__colsample_bytree': 0.5617255083597996, 'reg__min_child_samples': 12, 'reg__reg_alpha': 8.973412896307753, 'reg__reg_lambda': 24.24009428173029, 'reg__min_split_gain': 4.582435969307951, 'reg__path_smooth': 2.602609359762081, 'reg__min_child_weight': 3.272290226756703, 'reg__feature_fraction': 0.6044491794863425, 'reg__bagging_fraction': 0.5249991086266217, 'reg__bagging_freq': 2, 'reg__max_bin': 81, 'reg__min_data_per_group': 187}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.028500146956154604, 'reg__num_leaves': 51, 'reg__subsample': 0.6196447986657186, 'reg__colsample_bytree': 0.5771505489634249, 'reg__min_child_samples': 10, 'reg__reg_alpha': 9.532823043707634, 'reg__reg_lambda': 25.159613888695358, 'reg__min_split_gain': 4.2359225482602785, 'reg__path_smooth': 4.177150358462823, 'reg__min_child_weight':

5.727834520745972, 'reg__feature_fraction': 0.5805048020951218, 'reg__bagging_fraction': 0.5153630495662834, 'reg__bagging_freq': 2, 'reg__max_bin': 78, 'reg__min_data_per_group': 174}

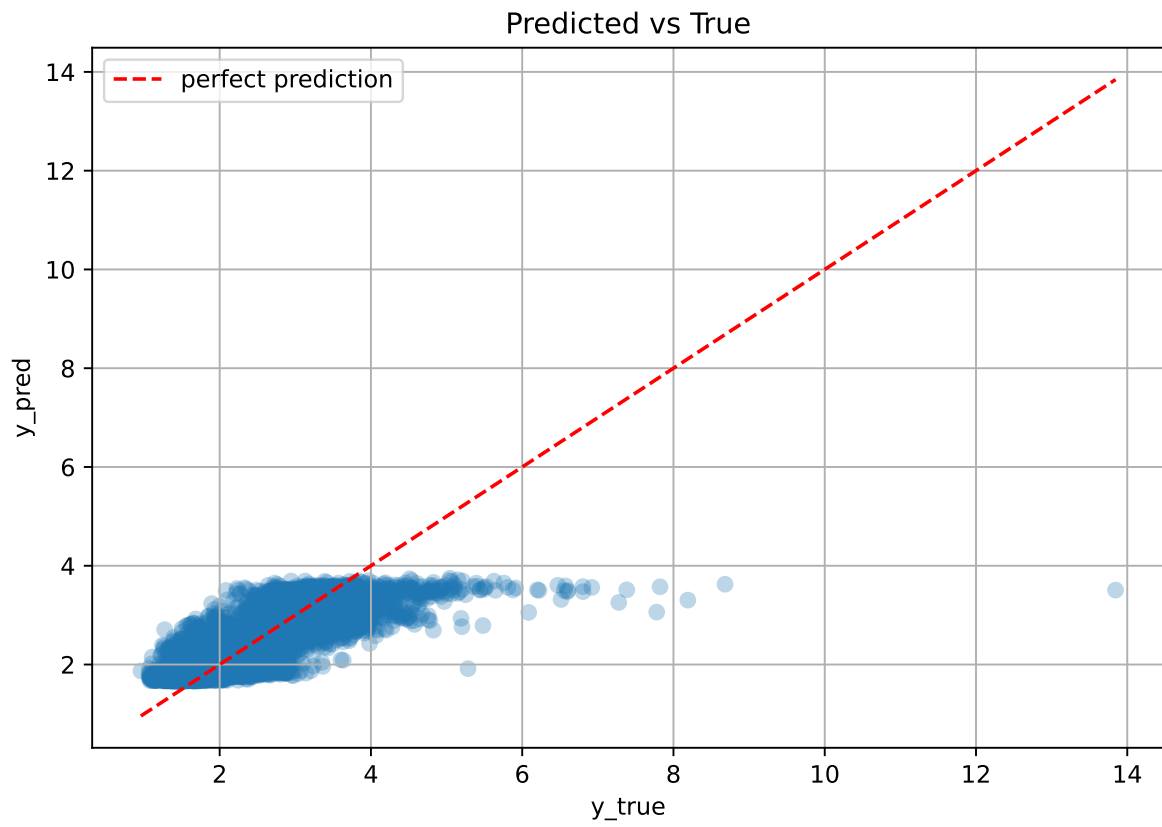
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.03296844213501357, 'reg__num_leaves': 46, 'reg__subsample': 0.6542895253760511, 'reg__colsample_bytree': 0.5676211600961134, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.807437278730811, 'reg__reg_lambda': 23.15583935996126, 'reg__min_split_gain': 4.510819385466352, 'reg__path_smooth': 2.5924759187572217, 'reg__min_child_weight': 2.747960552427293, 'reg__feature_fraction': 0.6077849004721779, 'reg__bagging_fraction': 0.5254827375621269, 'reg__bagging_freq': 2, 'reg__max_bin': 79, 'reg__min_data_per_group': 194}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.029180224230717627, 'reg__num_leaves': 51, 'reg__subsample': 0.6879732497666954, 'reg__colsample_bytree': 0.5814368585393763, 'reg__min_child_samples': 10, 'reg__reg_alpha': 9.859923951632386, 'reg__reg_lambda': 21.7012342326543, 'reg__min_split_gain': 3.8351530048759246, 'reg__path_smooth': 2.359365007794206, 'reg__min_child_weight': 7.818285691995614, 'reg__feature_fraction': 0.5878930358629059, 'reg__bagging_fraction': 0.5166473131851297, 'reg__bagging_freq': 2, 'reg__max_bin': 78, 'reg__min_data_per_group': 176}

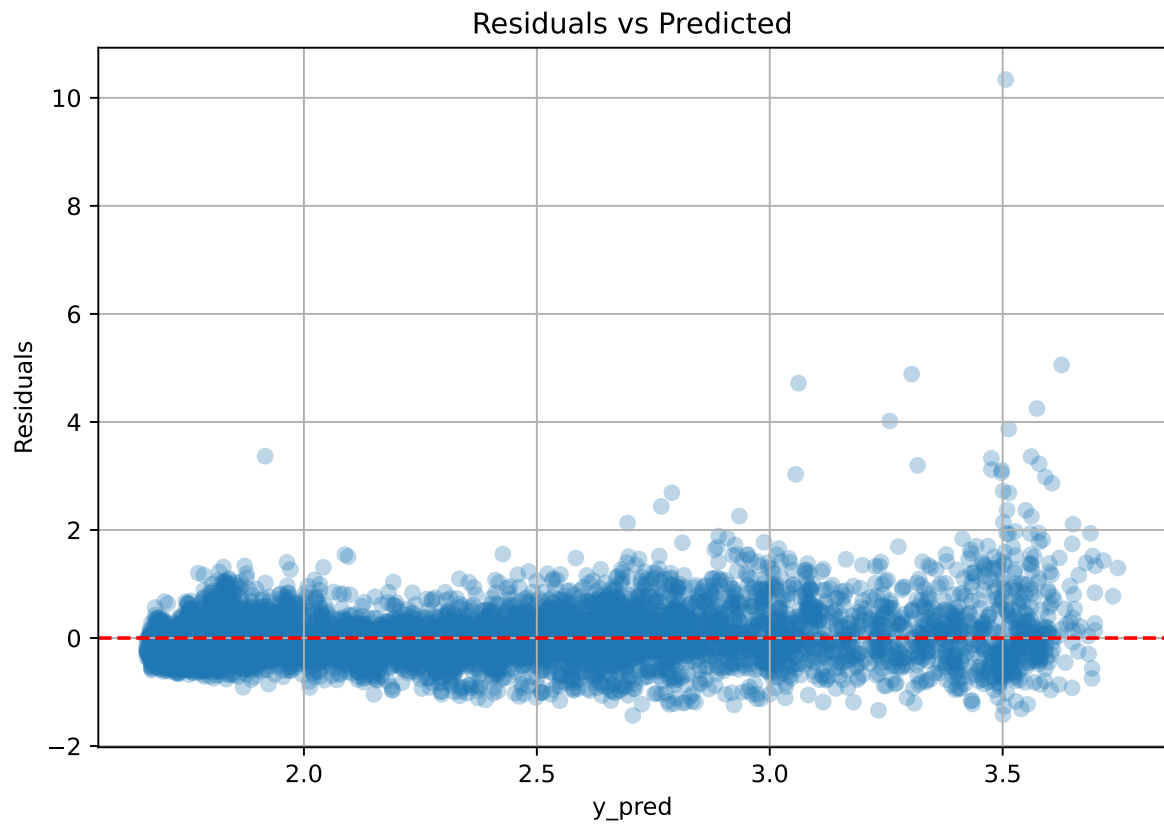
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.029813526140425234, 'reg__num_leaves': 46, 'reg__subsample': 0.5302131955934221, 'reg__colsample_bytree': 0.5840459778412279, 'reg__min_child_samples': 17, 'reg__reg_alpha': 9.034682711190364, 'reg__reg_lambda': 12.033914328958751, 'reg__min_split_gain': 4.441847001180282, 'reg__path_smooth': 3.2331109405156506, 'reg__min_child_weight': 0.002740679472525402, 'reg__feature_fraction': 0.5885957426415716, 'reg__bagging_fraction': 0.5238860560082883, 'reg__bagging_freq': 2, 'reg__max_bin': 79, 'reg__min_data_per_group': 190}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.019541658966775448, 'reg__num_leaves': 40, 'reg__subsample': 0.5549013456277746, 'reg__colsample_bytree': 0.5509663448470408, 'reg__min_child_samples': 25, 'reg__reg_alpha': 7.400361281948299, 'reg__reg_lambda': 22.73756246964094, 'reg__min_split_gain': 4.949645601865145, 'reg__path_smooth': 2.7533195249158253, 'reg__min_child_weight': 1.8236635205267482, 'reg__feature_fraction': 0.6312585464362617, 'reg__bagging_fraction': 0.5536727013373206, 'reg__bagging_freq': 3, 'reg__max_bin': 87, 'reg__min_data_per_group': 60}

3.5.2 Scatter



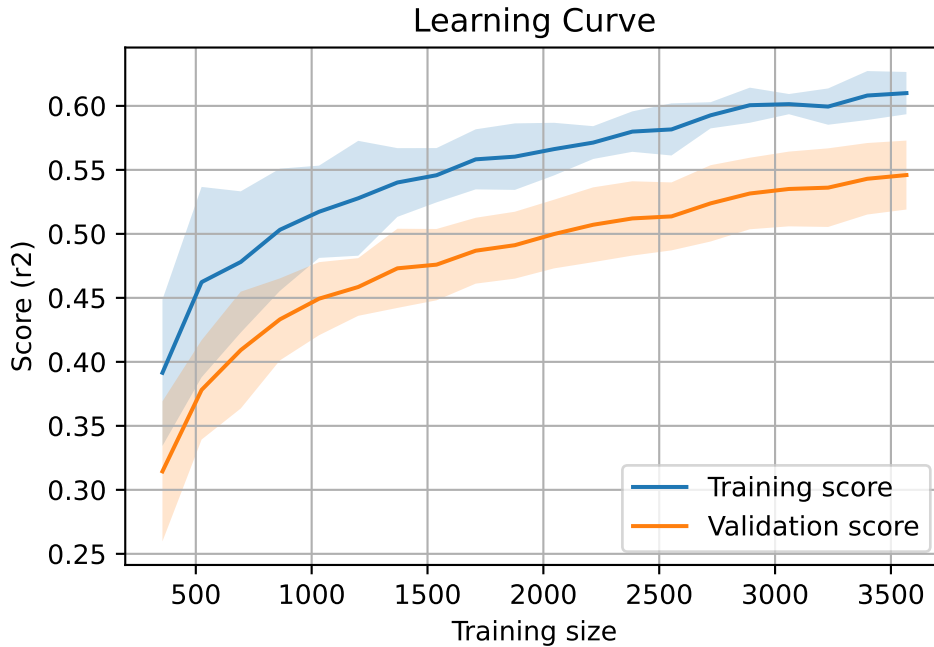
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7280, `Len(training)` : 5096,

`Len(training_real)` : 3568, `Len(test_of_training)` : 1528, `Len(test)` : 2184



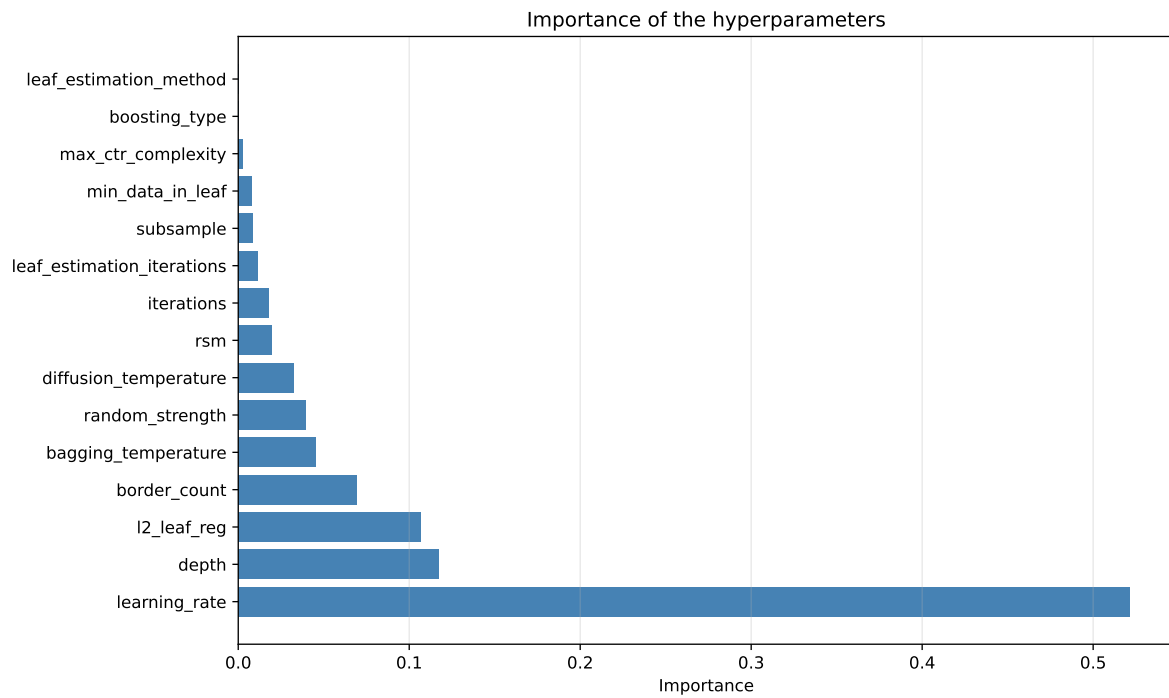
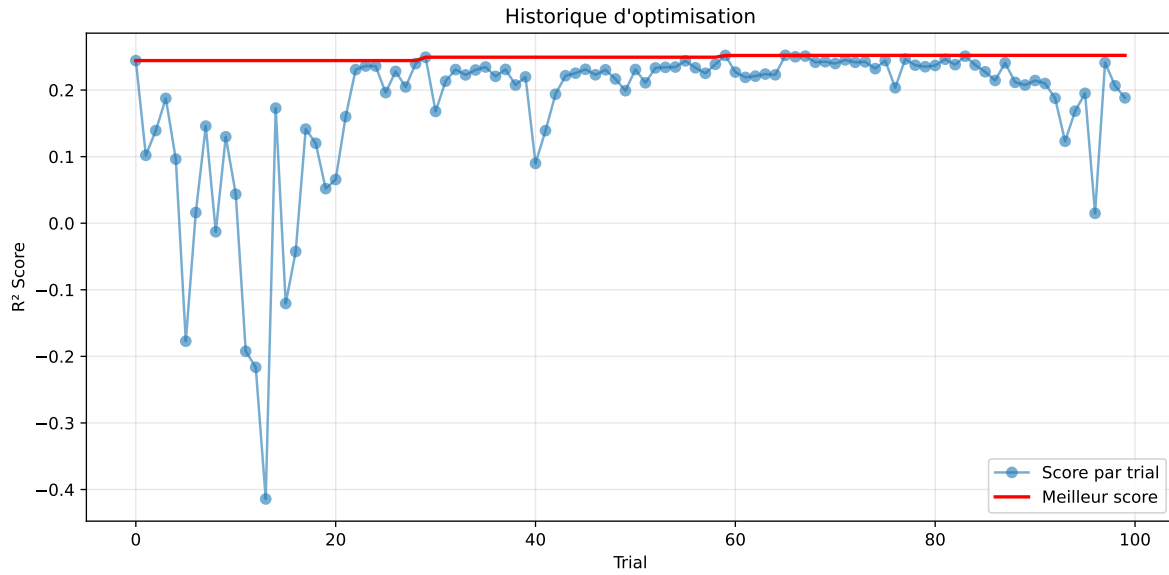
3.6 Model : CatBoost

3.6.1 Gridsearch

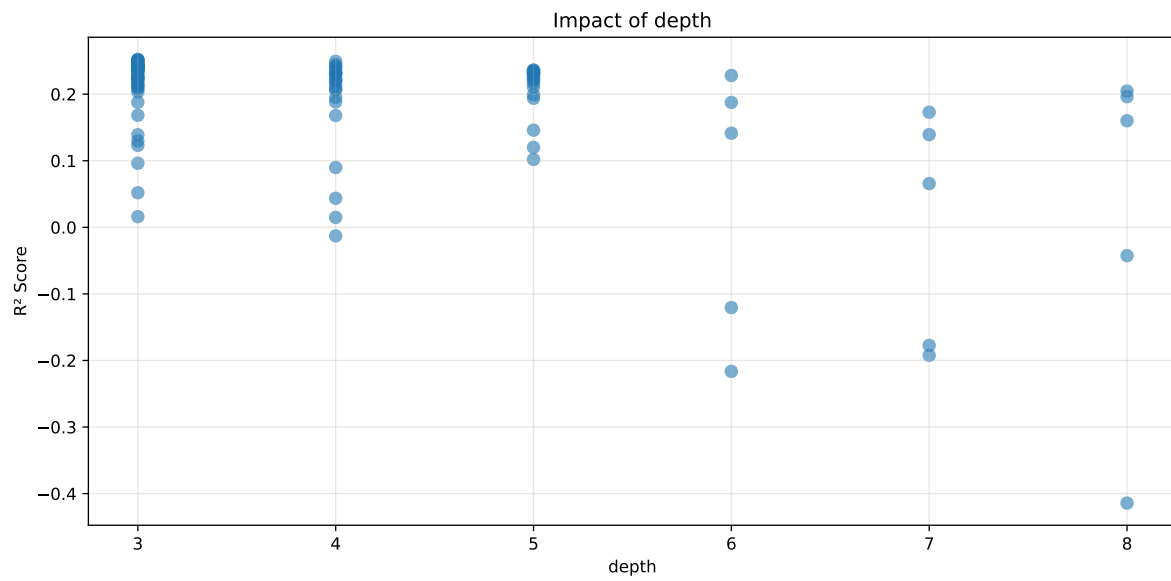
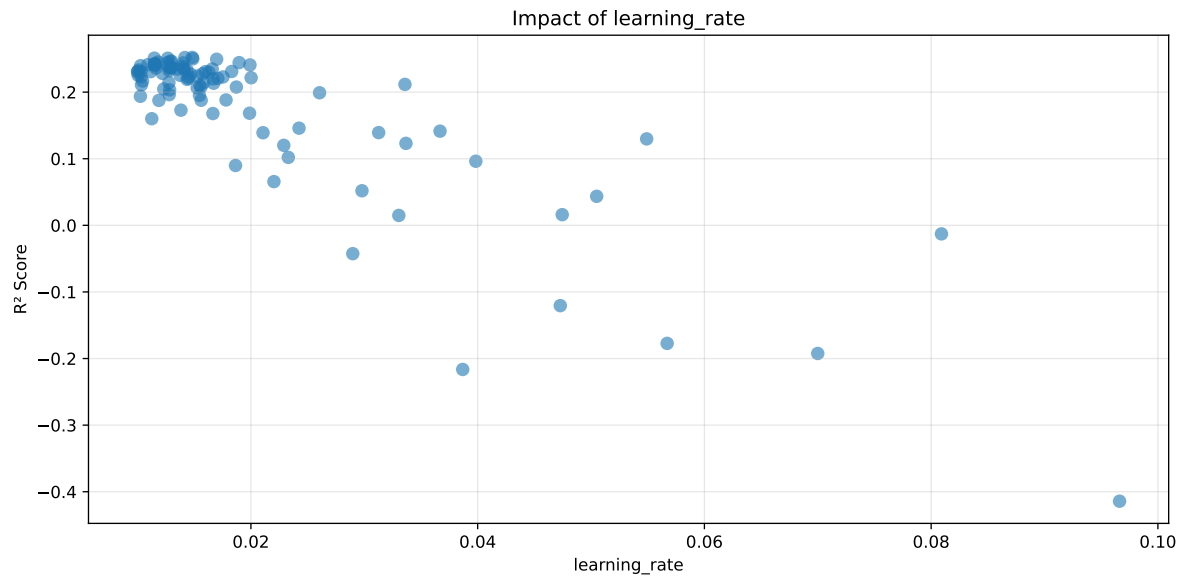
Distribution y_train vs y_test: y_train: mean=2.362, std=0.816, min=0.959, max=13.845
y_test: mean=2.359, std=0.781, min=1.062, max=7.822

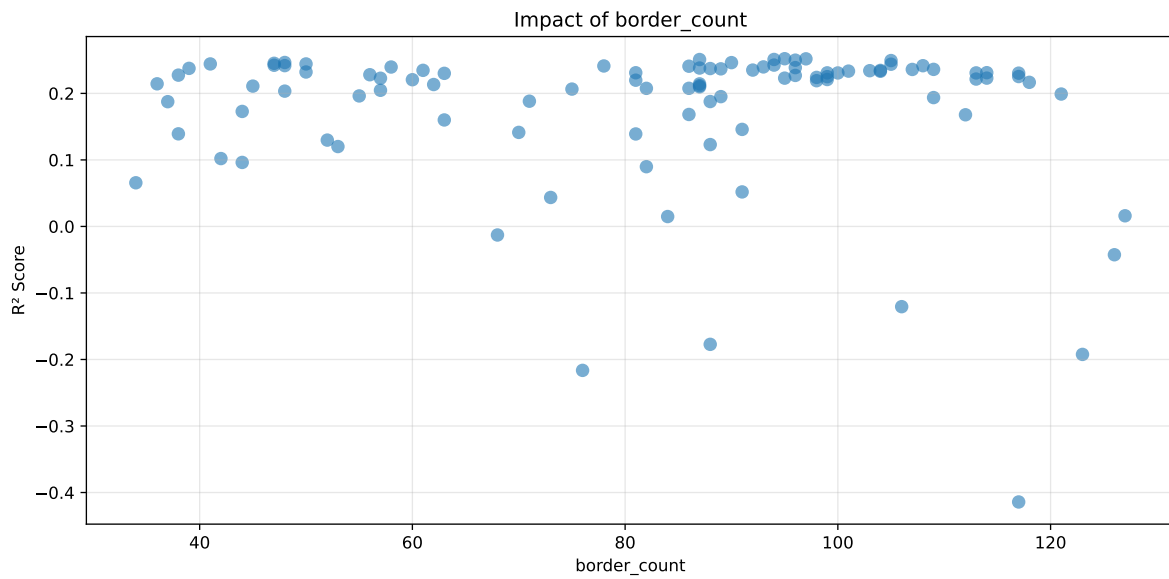
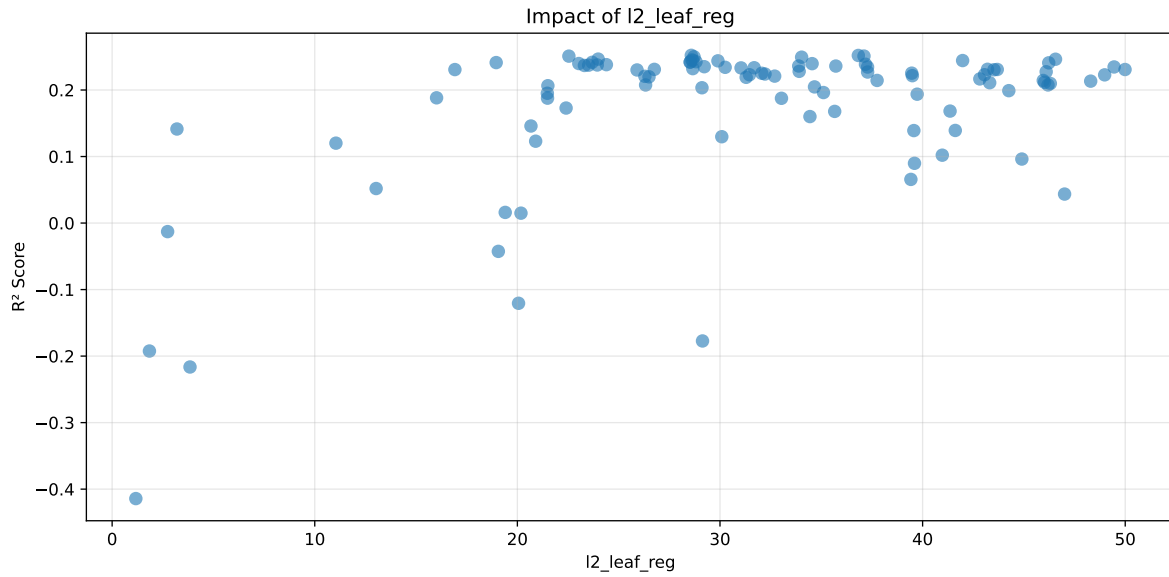
===== TOP Importance FEATURES =====

feature	importance
r_umidity	29.818098
r_temperature_st_24h_diff_last	20.744793
r_temperature	15.380103
r_umidity_st_24h_slope	7.485663
r_THI	5.616533
r_umidity_st_24h_diff_last	5.336227
r_umidity_st_24h_mean	4.136029
r_CO2	2.550630
r_NH3_st_24h_mean	2.334820
r_NH3	2.024045
r_umidity_st_24h_std	1.536260
r_temperature_st_24h_min	1.424881
r_NH3_st_24h_diff_last	1.032693
r_NH3_st_24h_slope	0.579224



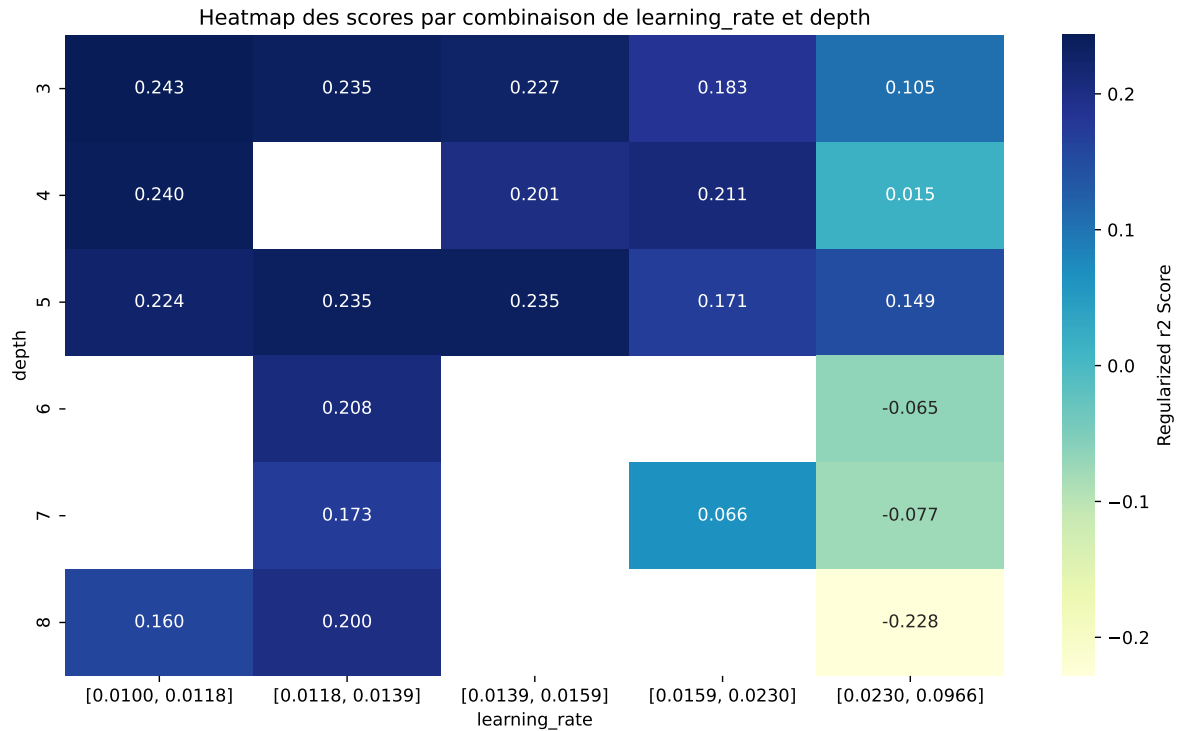
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'l2_leaf_reg', 'border_count']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5915	0.535	0.6084	0.0317	0.2521	0.6149	0.0799	1.1494	1.1057
2	0.6003	0.5422	0.6166	0.0313	0.2519	0.6192	0.077	1.142	1.1071
3	0.5831	0.5277	0.6015	0.0304	0.251	0.6054	0.0777	1.1472	1.1049
4	0.5548	0.5041	0.5748	0.0313	0.2509	0.5791	0.0749	1.1486	1.1005
5	0.5979	0.5399	0.614	0.0314	0.25	0.6175	0.0776	1.1436	1.1074
6	0.582	0.5266	0.6016	0.0317	0.2494	0.5988	0.0722	1.1372	1.1053
7	0.6207	0.5583	0.6326	0.0323	0.2465	0.6342	0.0759	1.1359	1.1117
8	0.5592	0.5071	0.5794	0.0301	0.2463	0.5826	0.0756	1.149	1.1029
9	0.5391	0.4901	0.5602	0.0288	0.2452	0.564	0.0738	1.1506	1.0999
10	0.609	0.5482	0.6203	0.0309	0.2443	0.6222	0.074	1.1351	1.1109
---	---	---	---	---	---	---	---	---	---
100	0.9982	0.7628	0.8341	0.0316	-0.4142	0.8486	0.0858	1.1125	1.3086

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01482732829383052, 'reg__l2_leaf_reg': 28.583882380643146, 'reg__bagging_temperature': 2.6125641385035347, 'reg__random_strength': 3.3870726410030434, 'reg__subsample': 0.8448394813916648, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8774035721231096, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 358.8919157666655}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01417587860326677, 'reg__l2_leaf_reg': 36.827188217999584, 'reg__bagging_temperature': 4.171902372254152, 'reg__random_strength': 3.3135387477353913, 'reg__subsample': 0.896538343493579, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8755293875883179, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3361.183177813045}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011480834793582239, 'reg__l2_leaf_reg': 37.106593964851825, 'reg__bagging_temperature': 3.961486747753457, 'reg__random_strength': 1.6825208358292005, 'reg__subsample': 0.8563517957948049, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8813806885492579, 'reg__border_count': 94, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 243.74297150006944}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.012668584207182048, 'reg__l2_leaf_reg': 22.541779466849647, 'reg__bagging_temperature': 2.2479145264849985, 'reg__random_strength': 3.656061201476221, 'reg__subsample': 0.8605538001597045, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8597514943116563, 'reg__border_count': 87, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4288.22671486353}

Rank 5: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01487989808451933, 'reg__l2_leaf_reg': 28.71940332364422, 'reg__bagging_temperature': 3.9736106491391148, 'reg__random_strength': 3.2969960708119377, 'reg__subsample': 0.8543643226135207, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8777139044210746, 'reg__border_count': 96, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 397.164557596346}

Rank 6: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.0169788508104617, 'reg__l2_leaf_reg': 34.02856194250673, 'reg__bagging_temperature': 4.627897587638886, 'reg__random_strength': 2.4587161467122867, 'reg__subsample': 0.5826641141987464, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8861798928084389, 'reg__border_count': 105, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 89.55943377906046}

Rank 7: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.012891253048838808, 'reg__l2_leaf_reg': 23.988545261234272, 'reg__bagging_temperature': 2.304767713840155, 'reg__random_strength': 3.4790613800934587, 'reg__subsample': 0.8515344075080759,

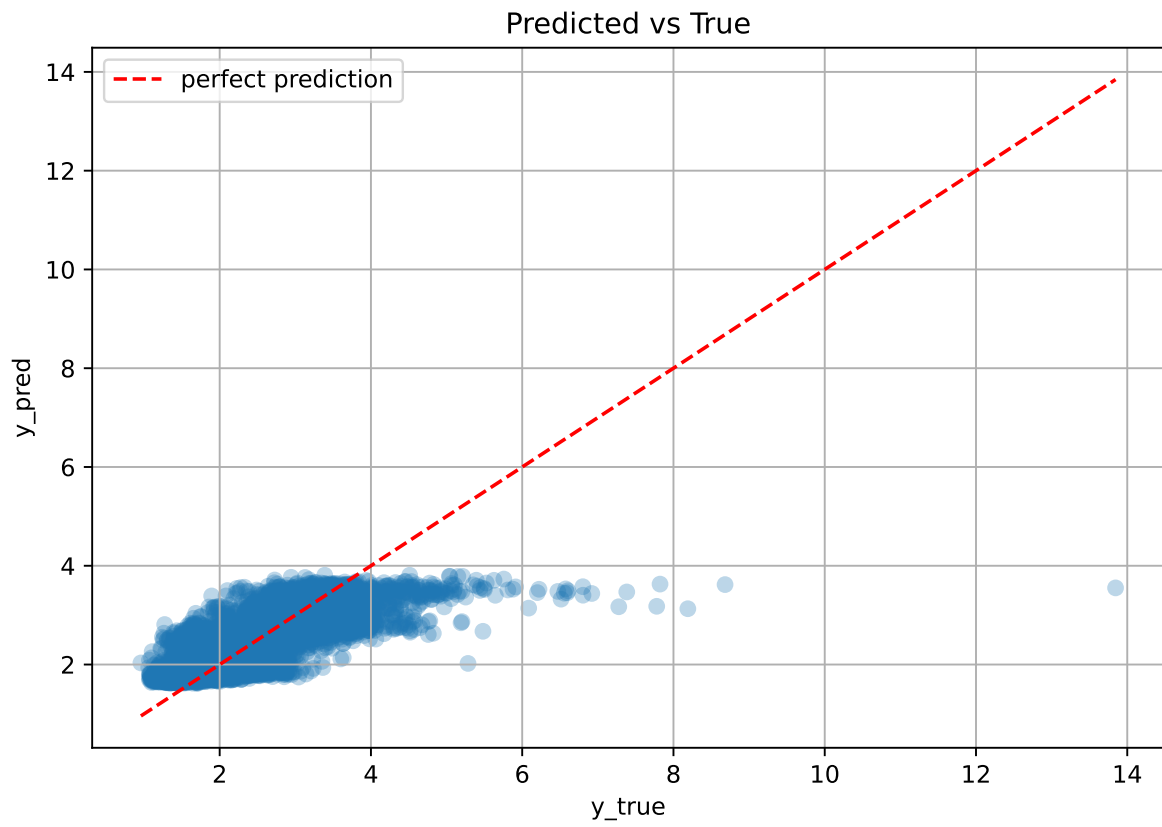
'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6682760853204064, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 302.3801602844547}

Rank 8: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.013001451886292637, 'reg__l2_leaf_reg': 46.559780852672986, 'reg__bagging_temperature': 2.275889762035788, 'reg__random_strength': 3.1768776012410105, 'reg__subsample': 0.8591801423124653, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8605119111959647, 'reg__border_count': 90, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4149.420655645891}

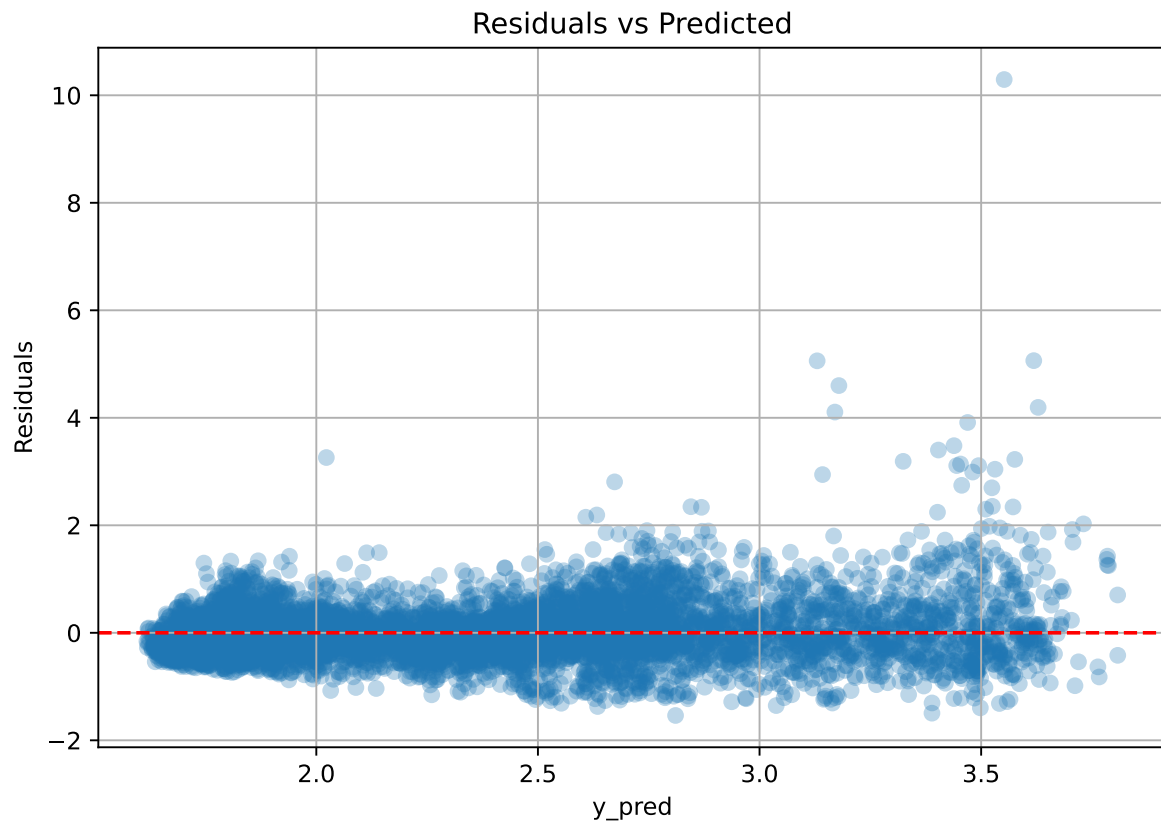
Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01179864989665897, 'reg__l2_leaf_reg': 28.614458866978012, 'reg__bagging_temperature': 2.6647192044066603, 'reg__random_strength': 3.2332036737923486, 'reg__subsample': 0.7944245428840483, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8783733739864299, 'reg__border_count': 47, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 283.959901672089}

Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.018960766950291297, 'reg__l2_leaf_reg': 41.97443868028249, 'reg__bagging_temperature': 3.7272939652105466, 'reg__random_strength': 3.0845568430788513, 'reg__subsample': 0.5579402016783738, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8532875970278734, 'reg__border_count': 50, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 145.41813867072872}

3.6.2 Scatter



3.6.3 Scatter residuals



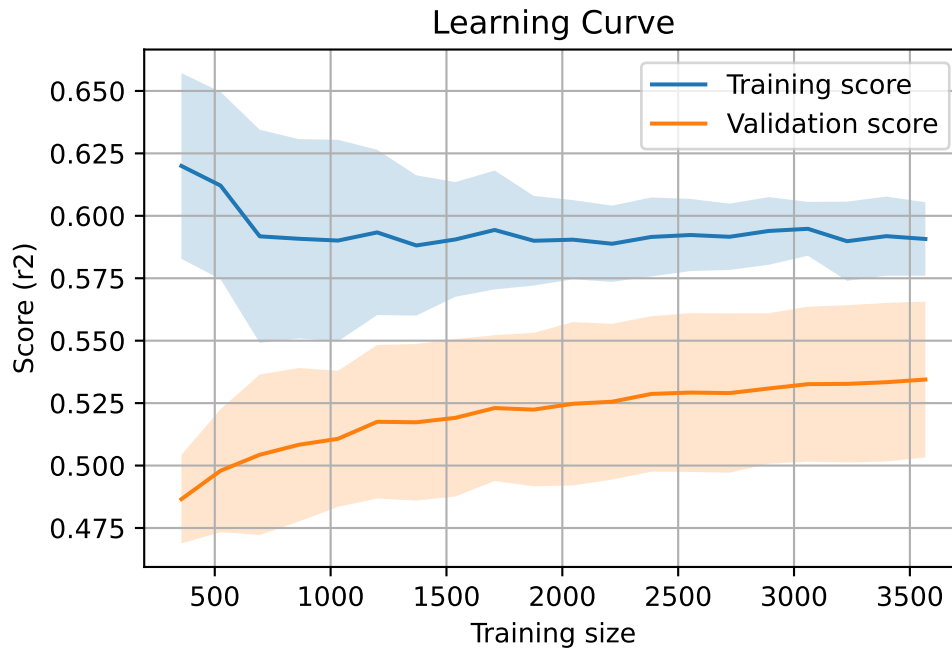
3.6.4 Learning curve

`Len(group)` : 7280, `Len(training)` : 5096,

`Len(training_real)` : 3568, `Len(test_of_training)` : 1528, `Len(test)` : 2184

C:\Users\titou\anaconda3\Lib\site-packages\joblib\externals\loky\process_executor.py:782: Use

A worker stopped while some jobs were given to the executor. This can be caused by a too sho



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.527	0.481	0.03	0.543	0.543	0.062	1.129	1.095	nan
LGBM	0.605	0.542	0.03	0.616	0.639	0.098	1.18	1.116	0.2278
CatBoost	0.592	0.535	0.032	0.608	0.615	0.08	1.149	1.106	0.2521

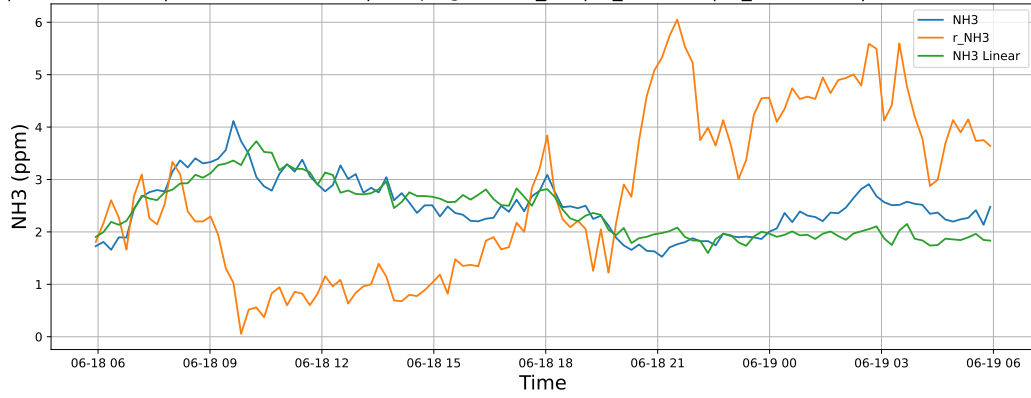
4 Plot

4.1 Standalone plots

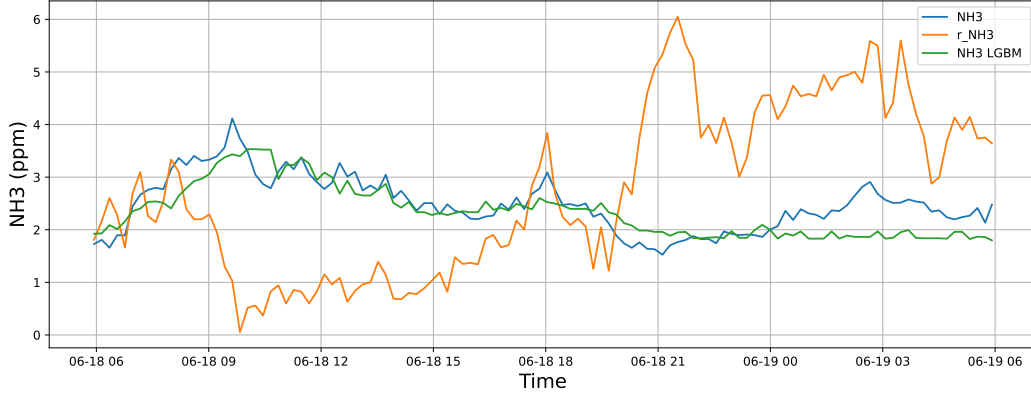
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

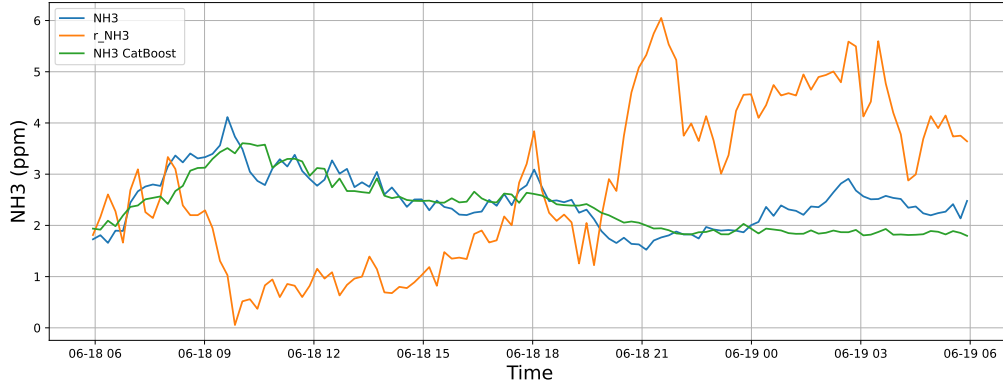
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

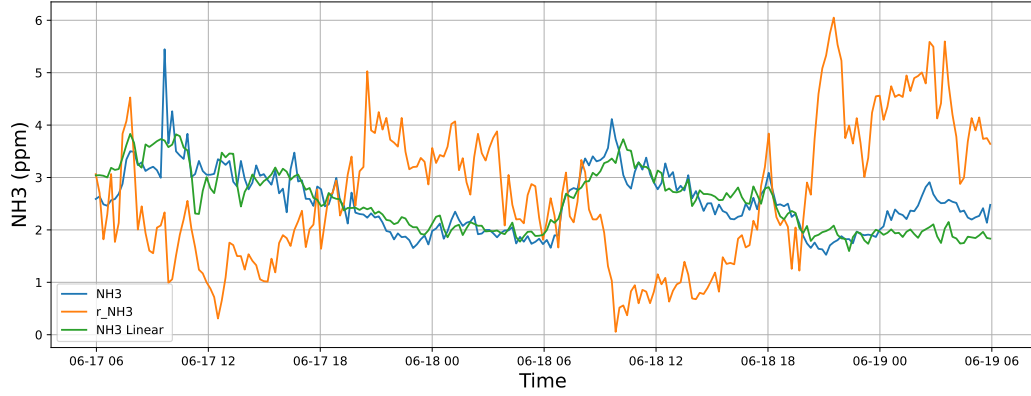


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

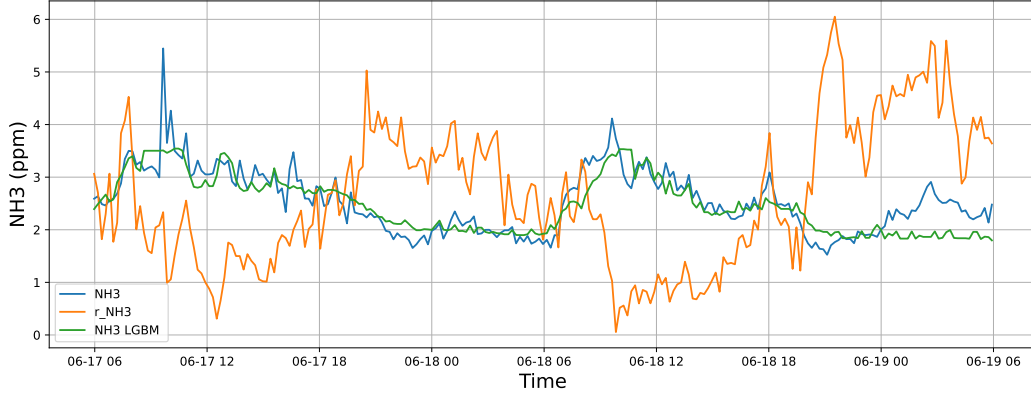


4.1.1.2 Time window : 48

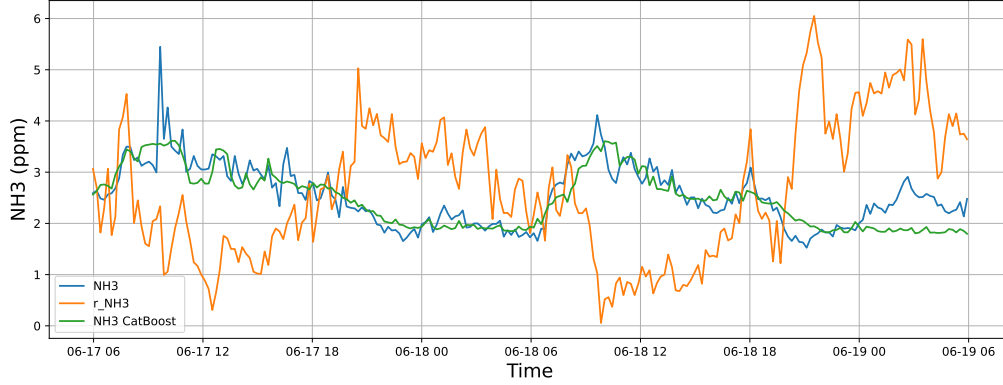
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

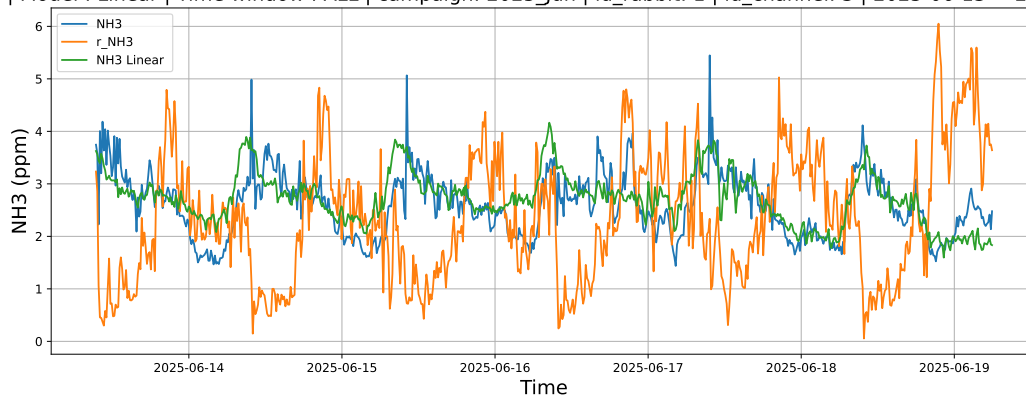


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

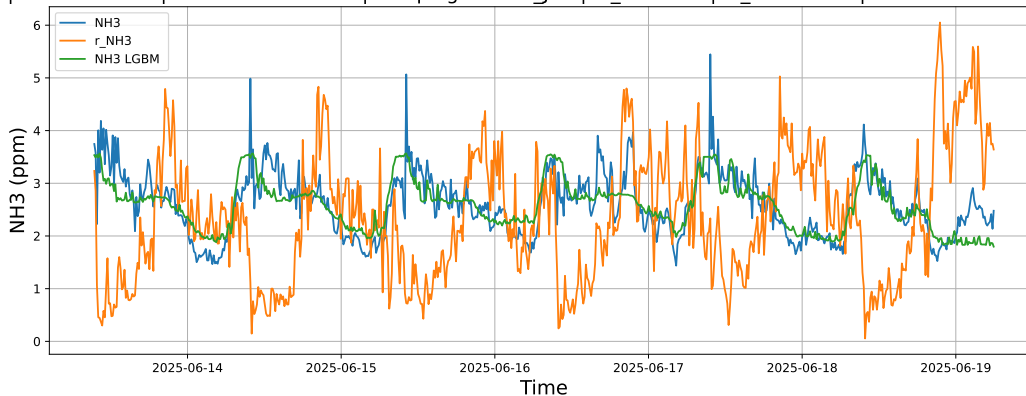


4.1.1.3 Time window : ALL

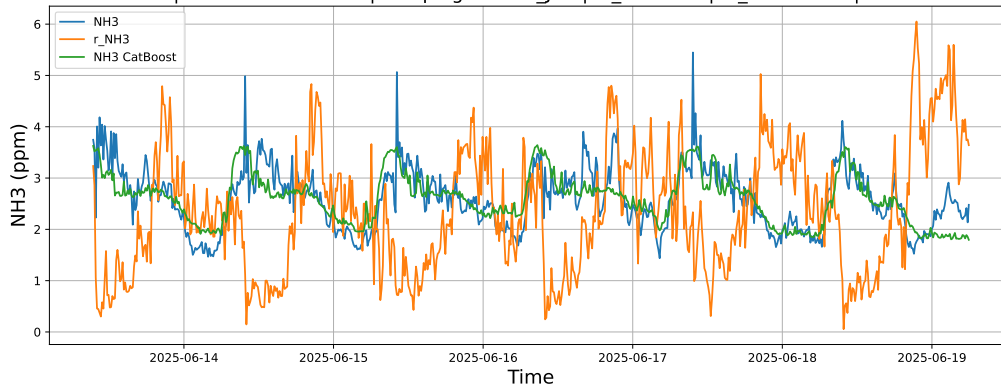
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



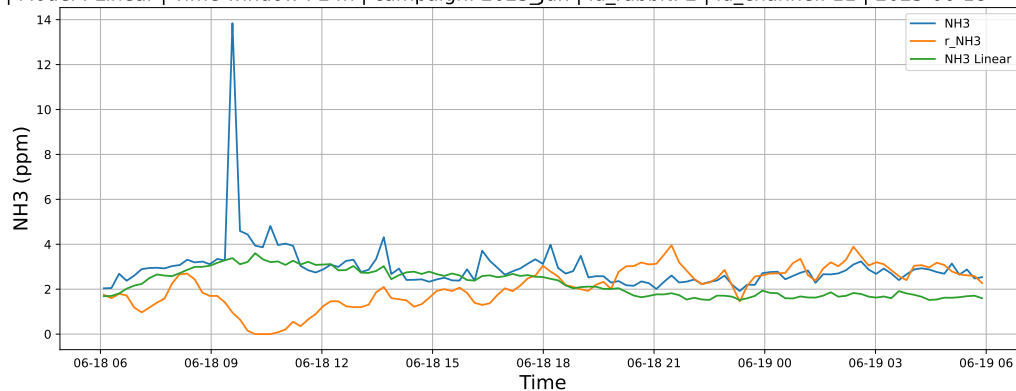
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



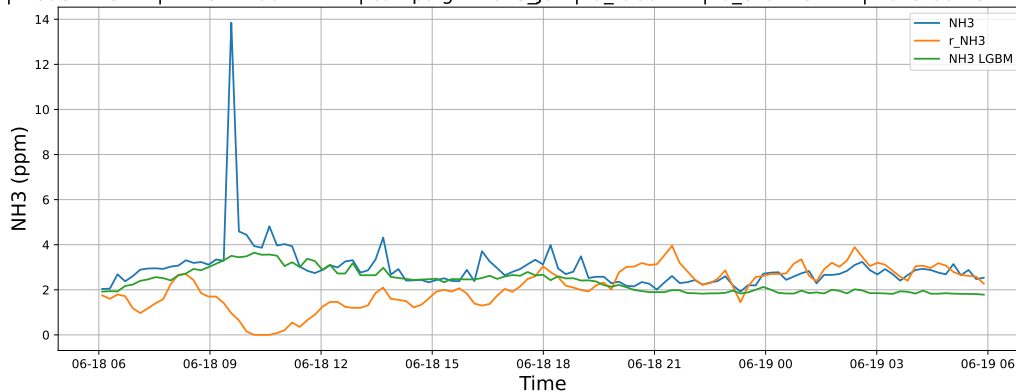
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

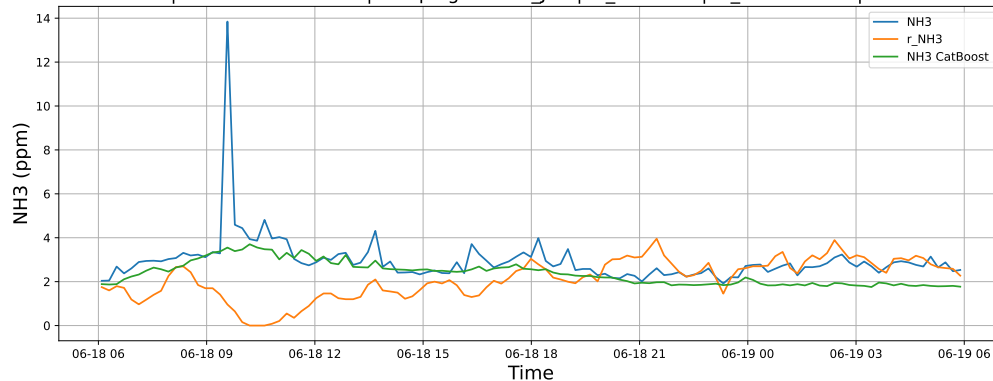
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

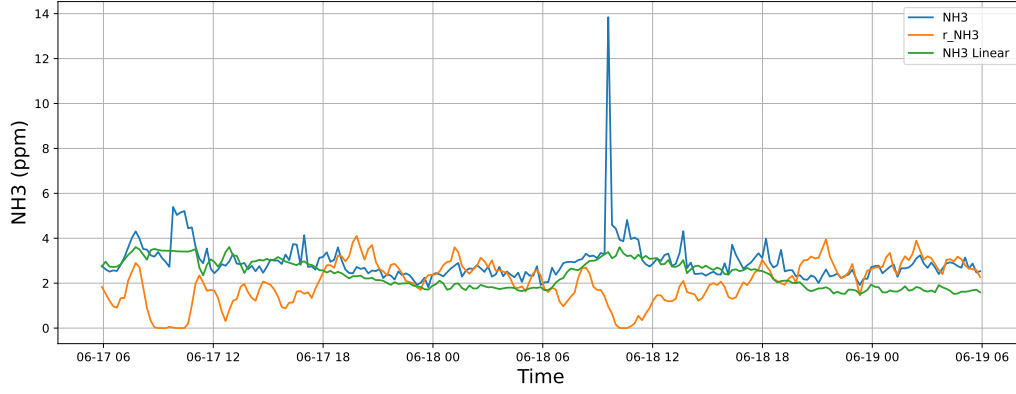


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

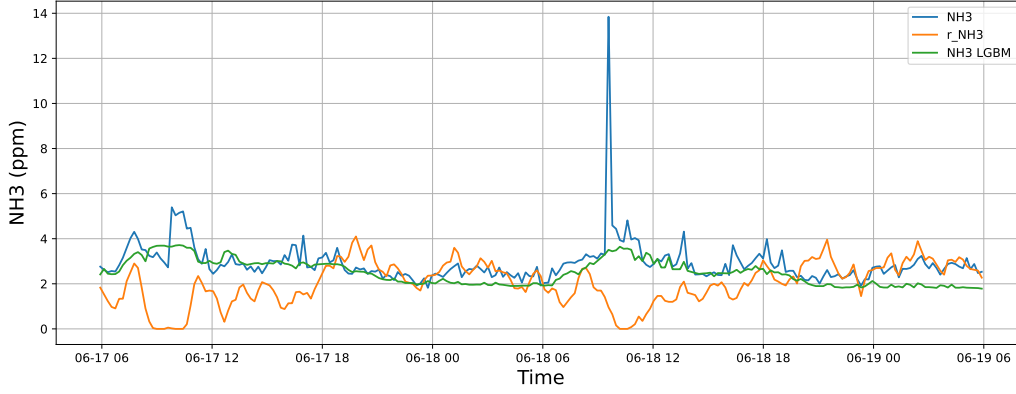


4.1.2.2 Time window : 48

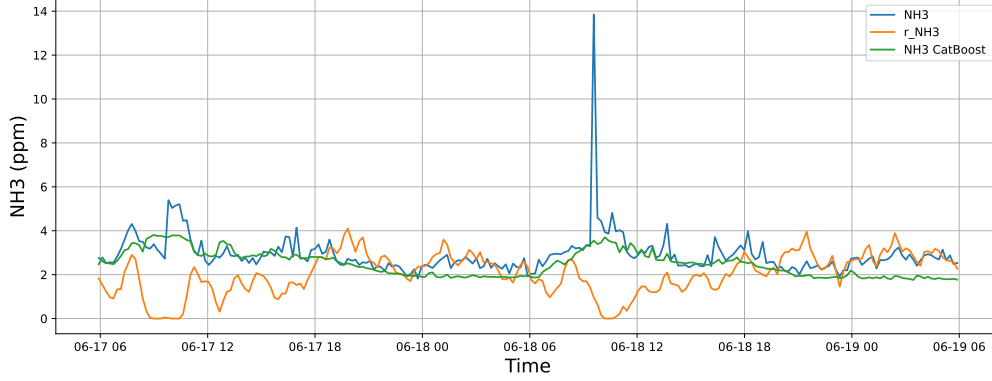
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

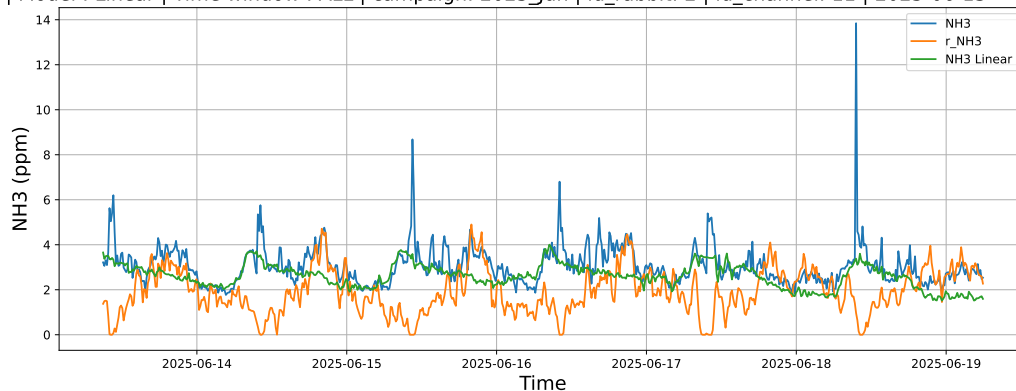


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

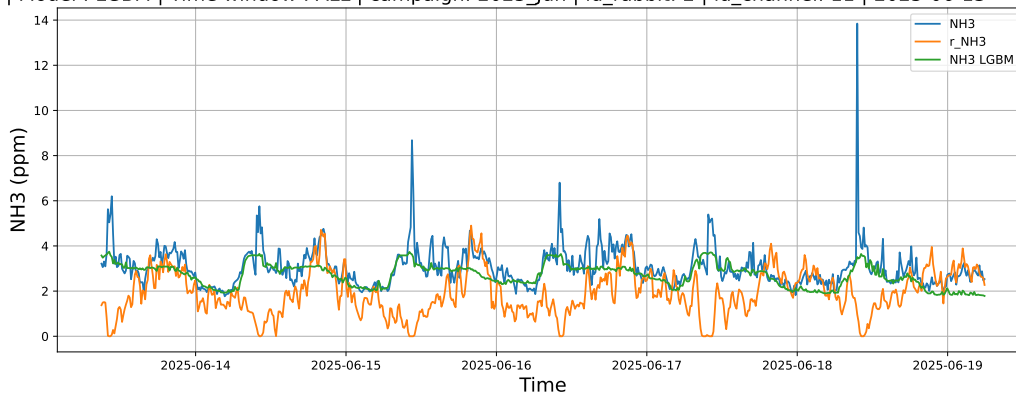


4.1.2.3 Time window : ALL

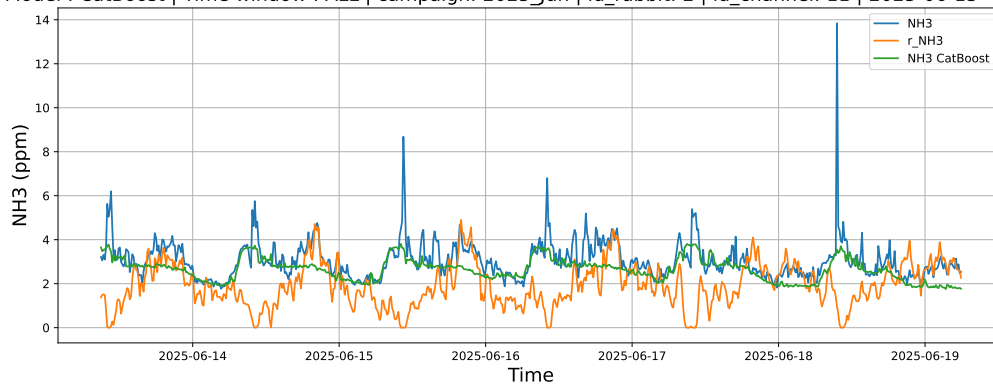
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



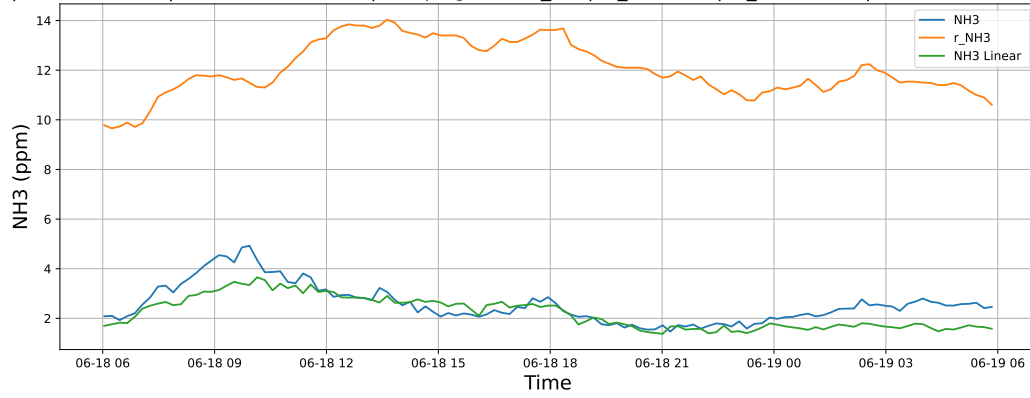
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



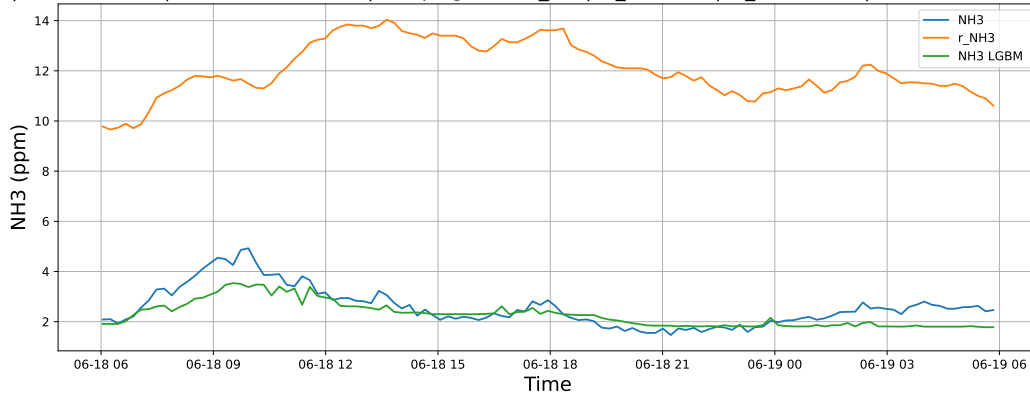
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

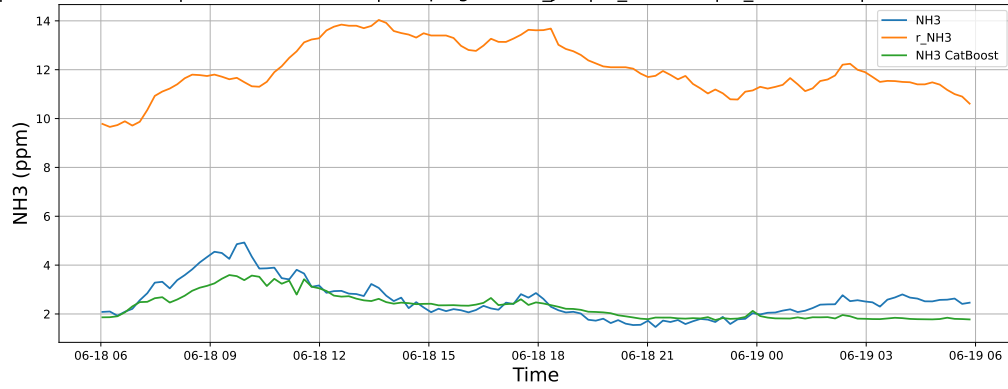
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

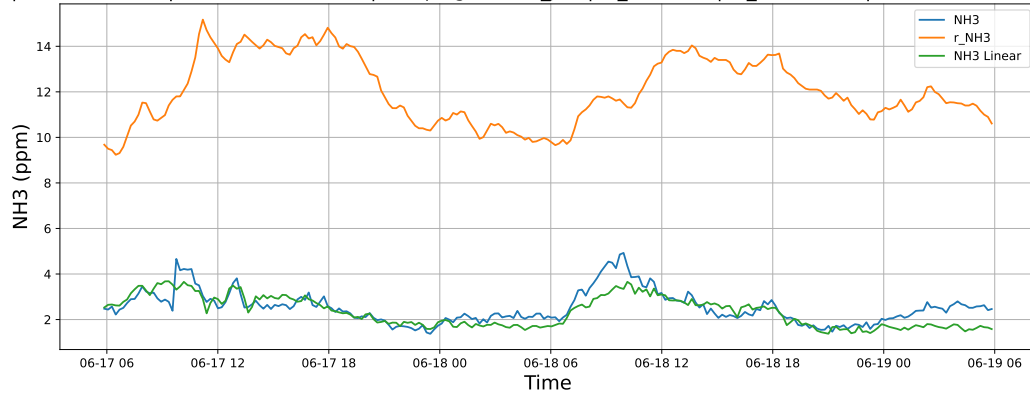


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

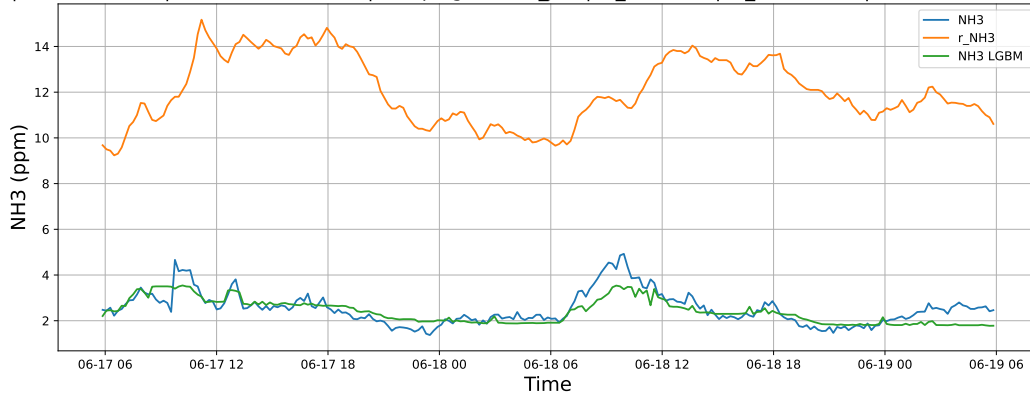


4.1.3.2 Time window : 48

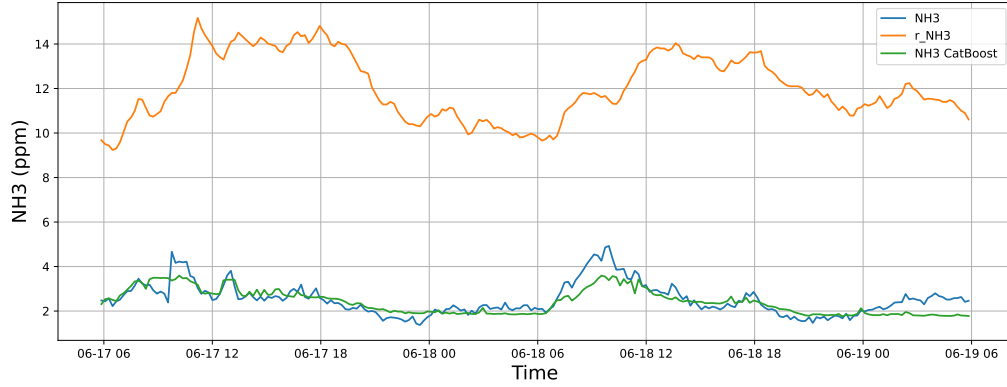
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

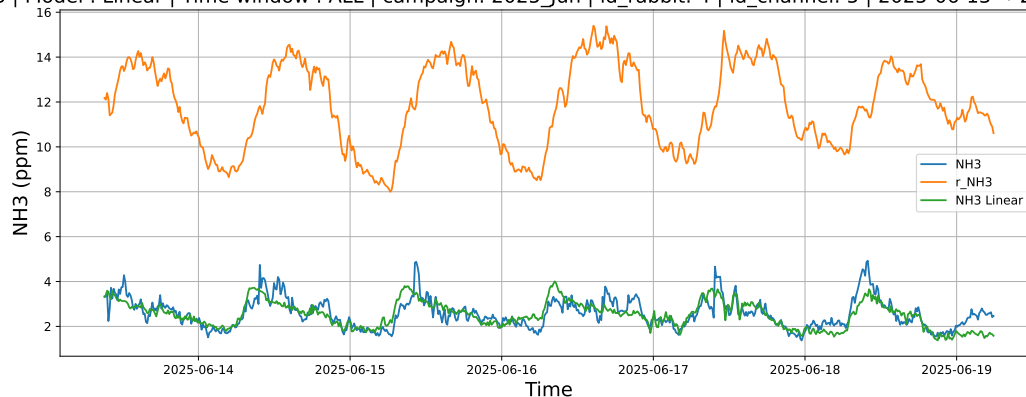


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

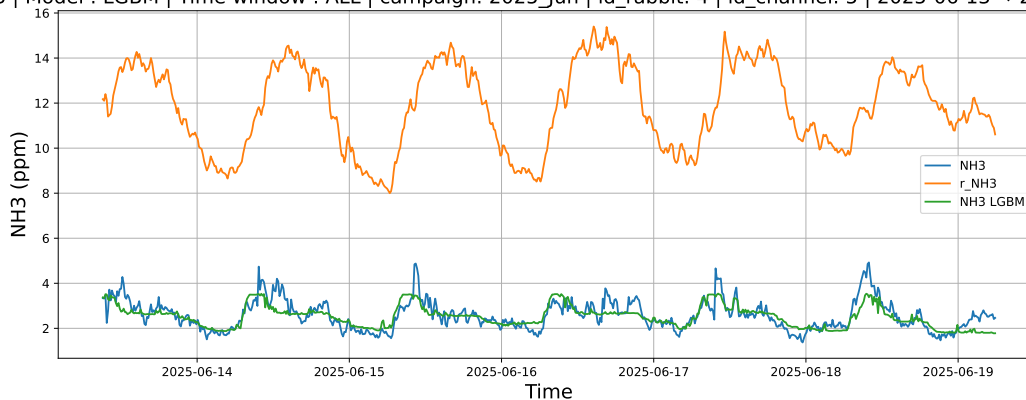


4.1.3.3 Time window : ALL

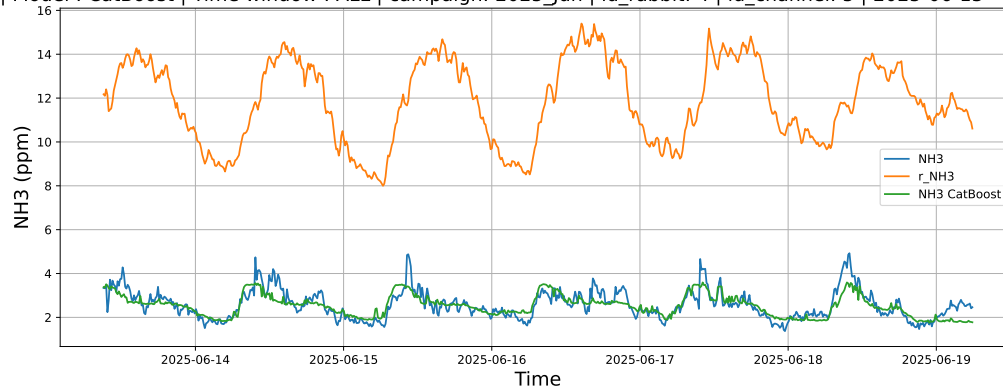
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



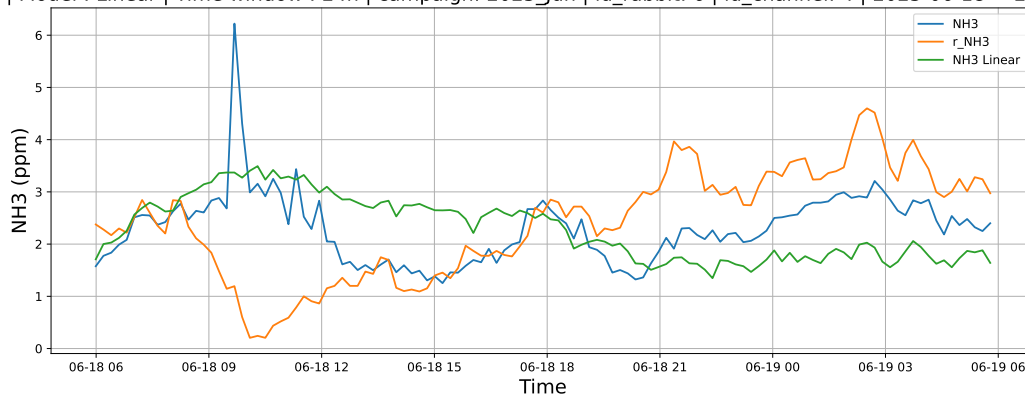
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



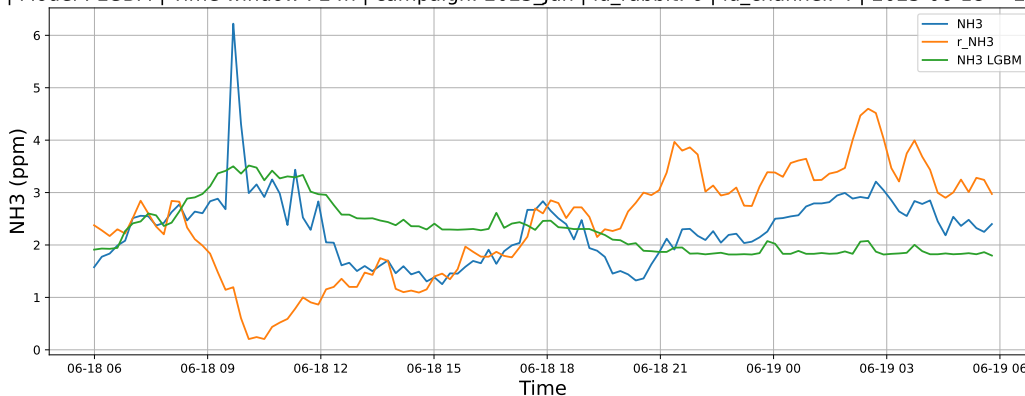
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

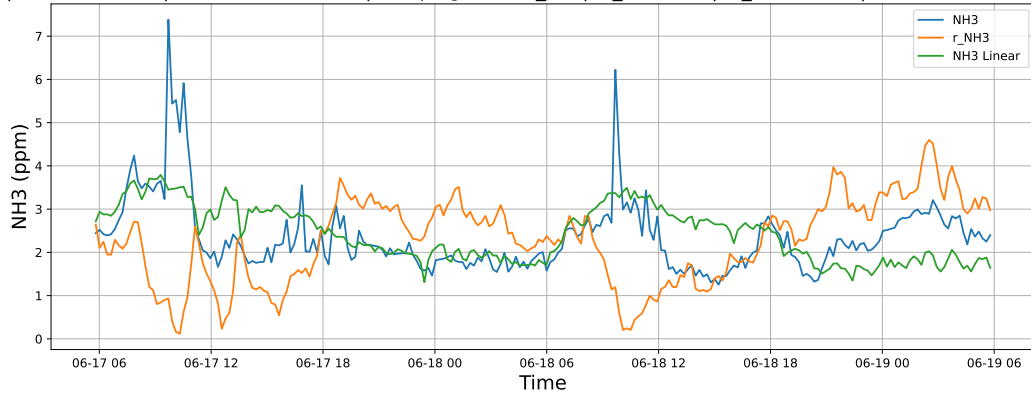


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

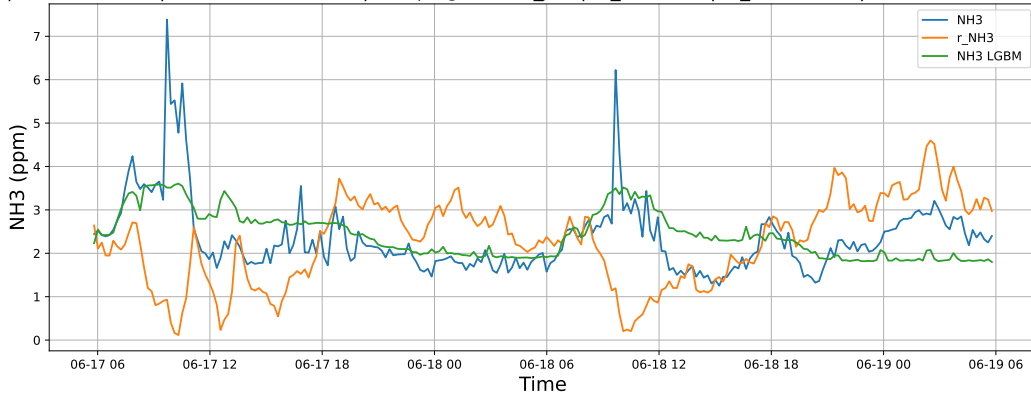


4.1.4.2 Time window : 48

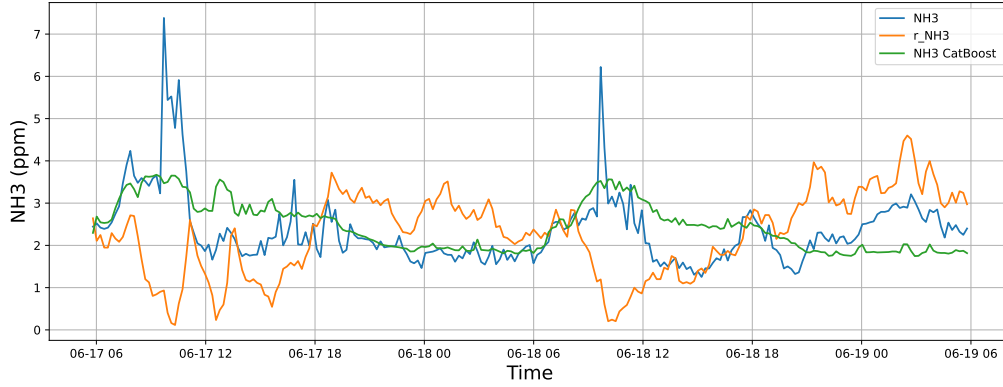
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

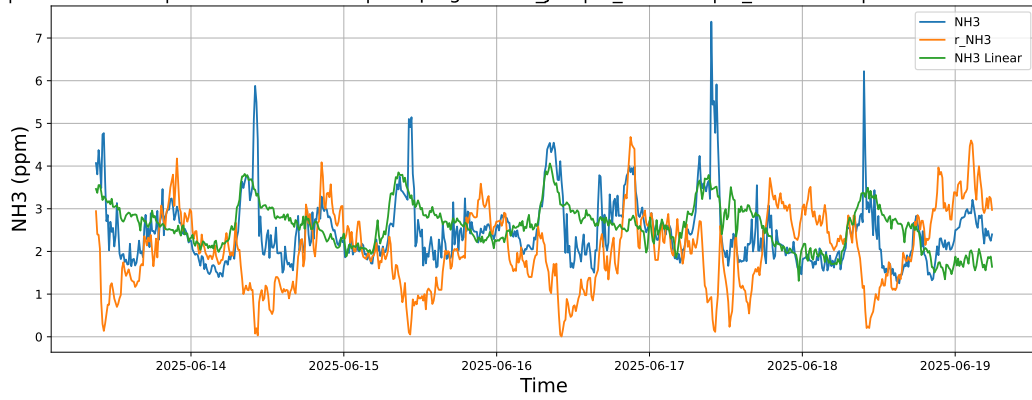


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

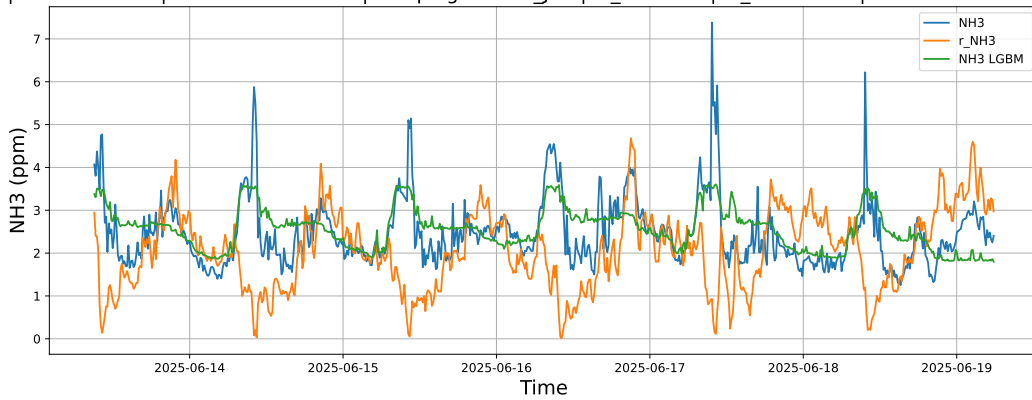


4.1.4.3 Time window : ALL

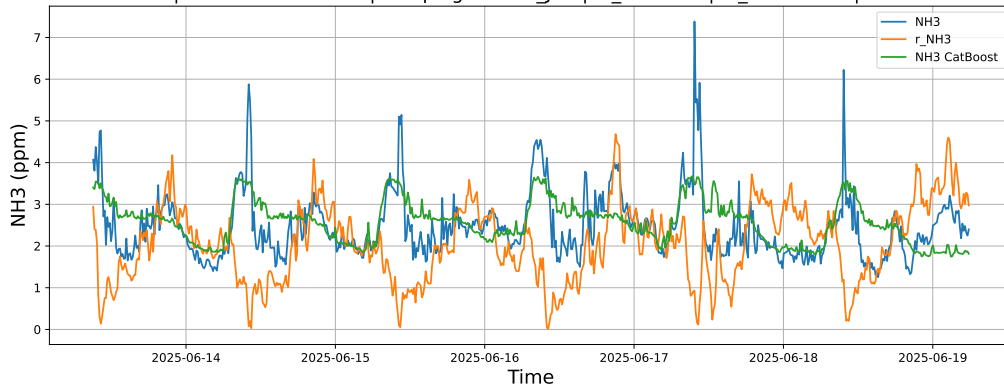
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



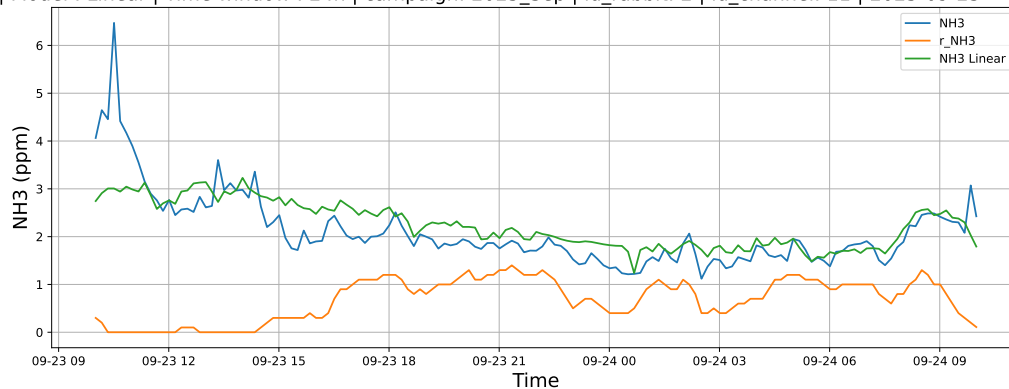
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



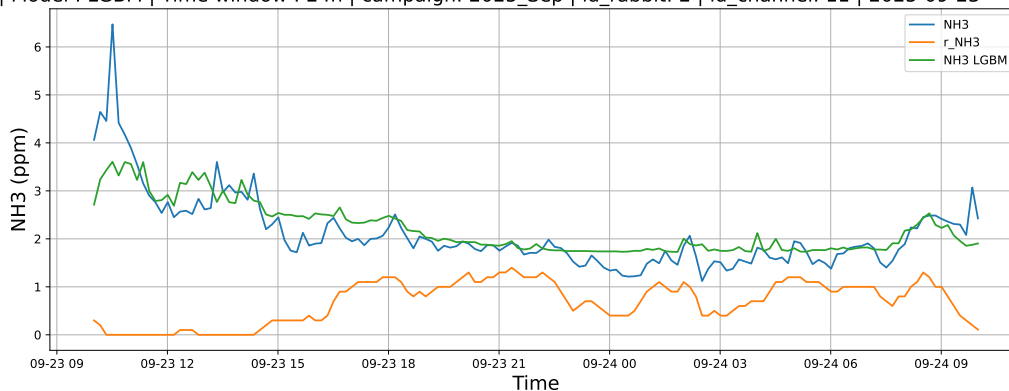
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

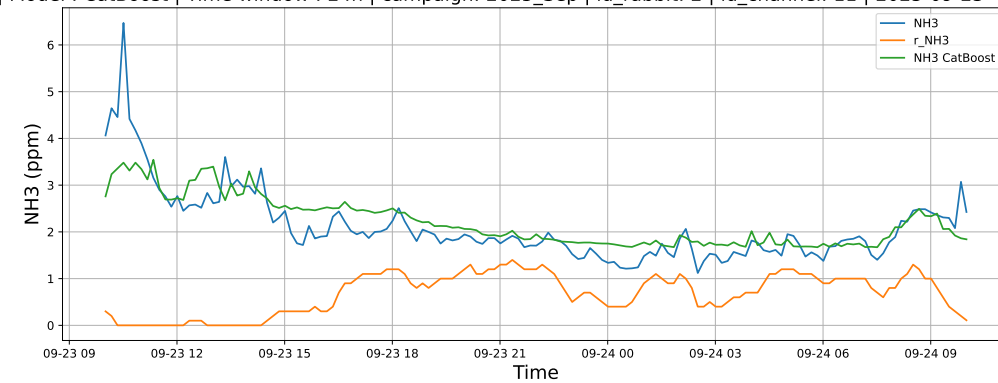
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

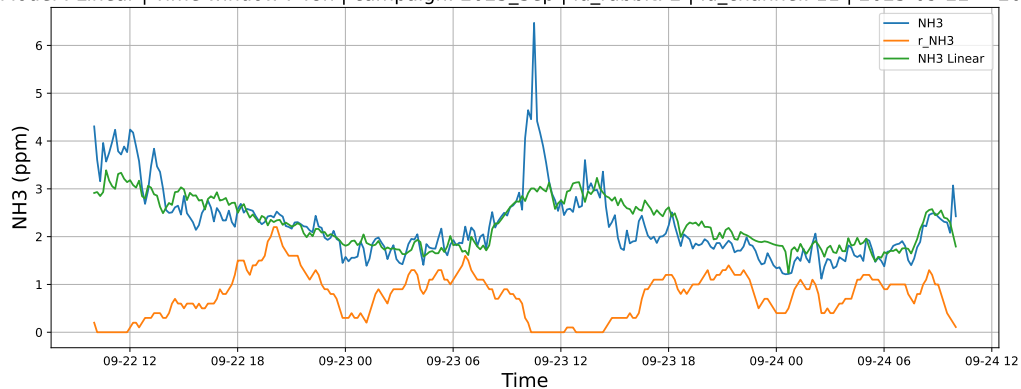


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

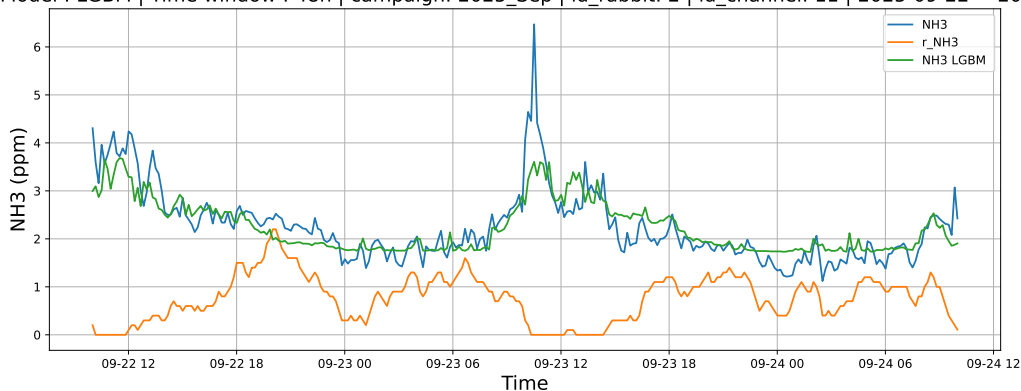


4.1.5.2 Time window : 48

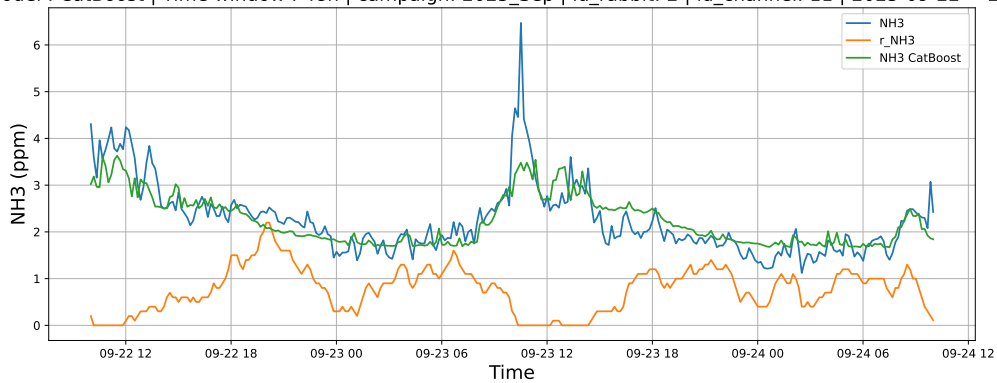
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

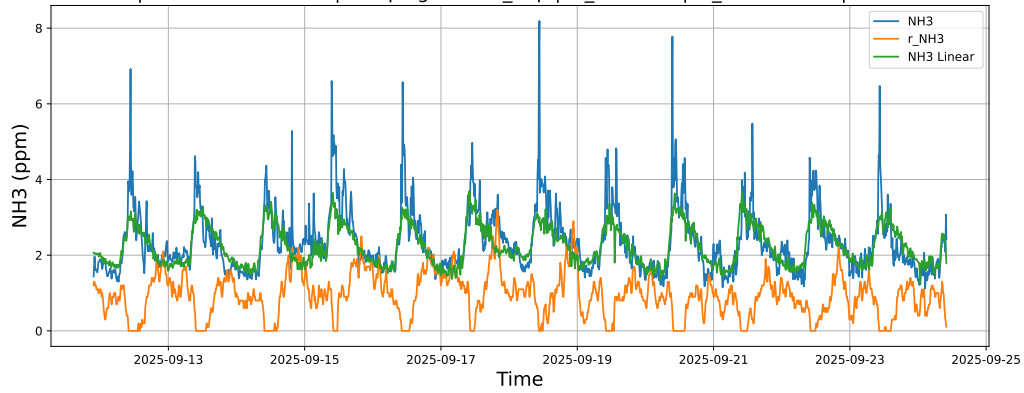


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

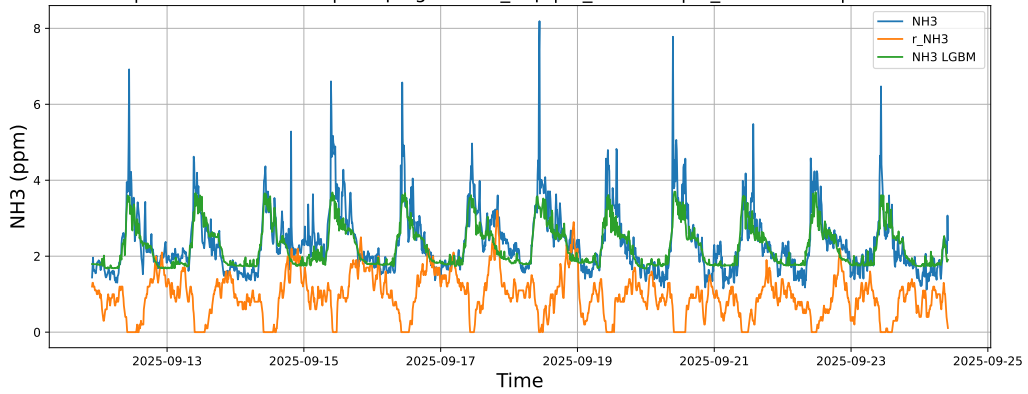


4.1.5.3 Time window : ALL

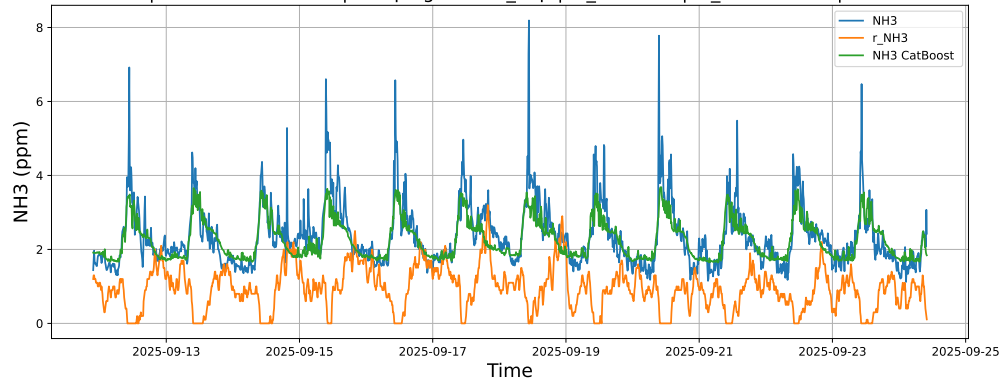
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



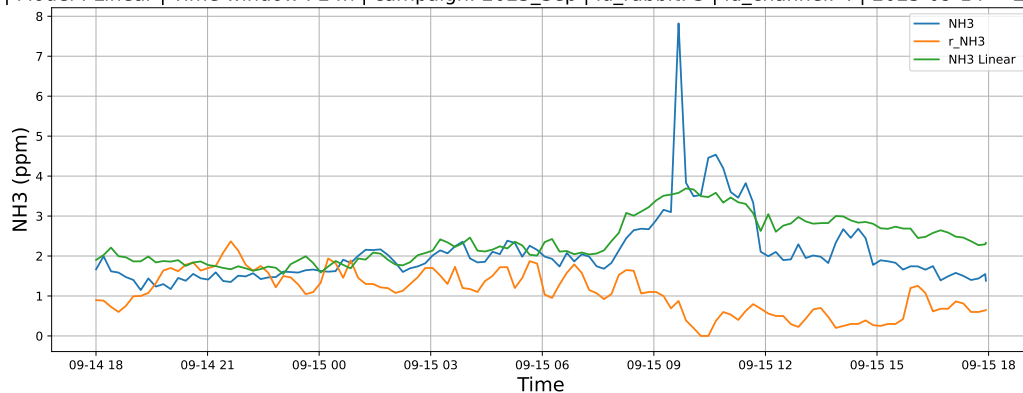
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



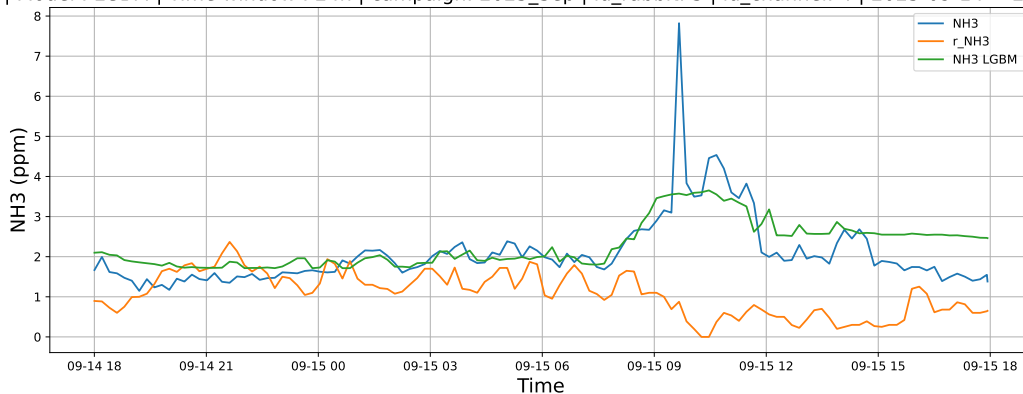
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

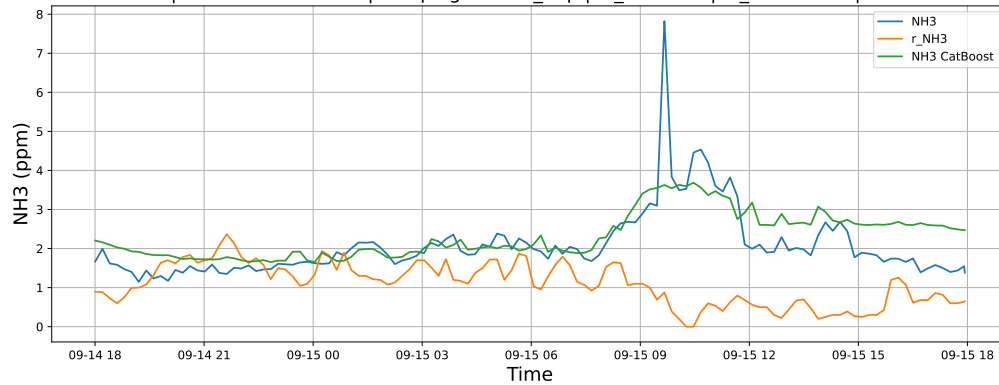
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

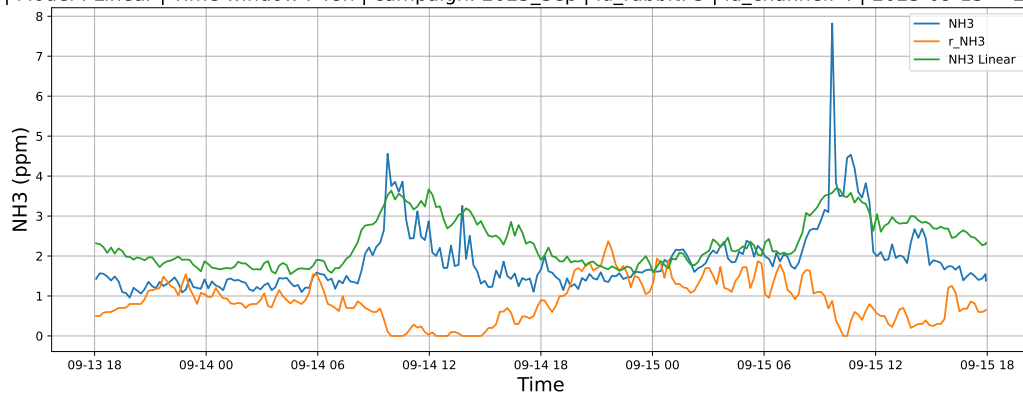


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

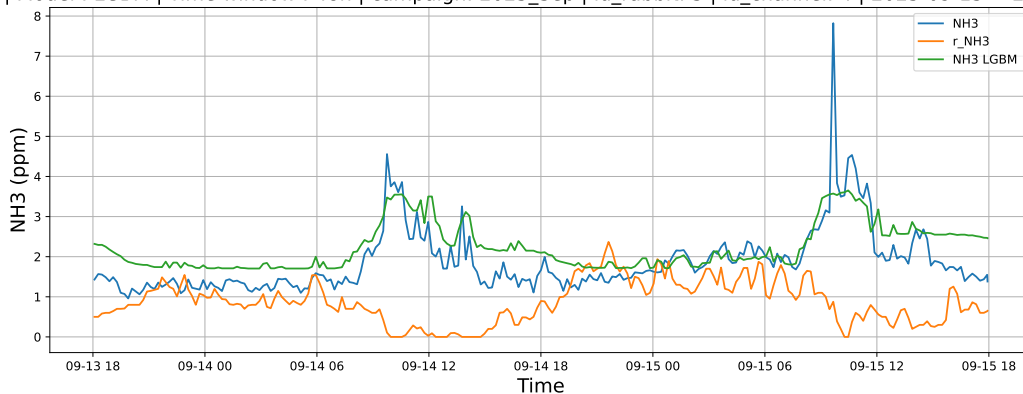


4.1.6.2 Time window : 48

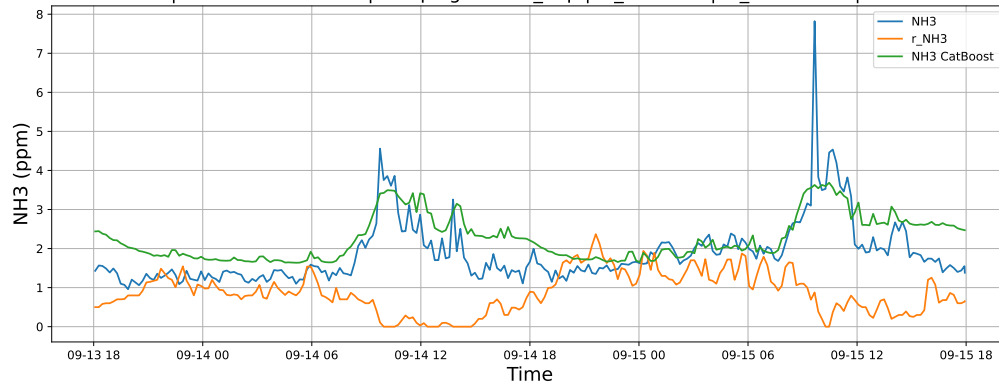
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

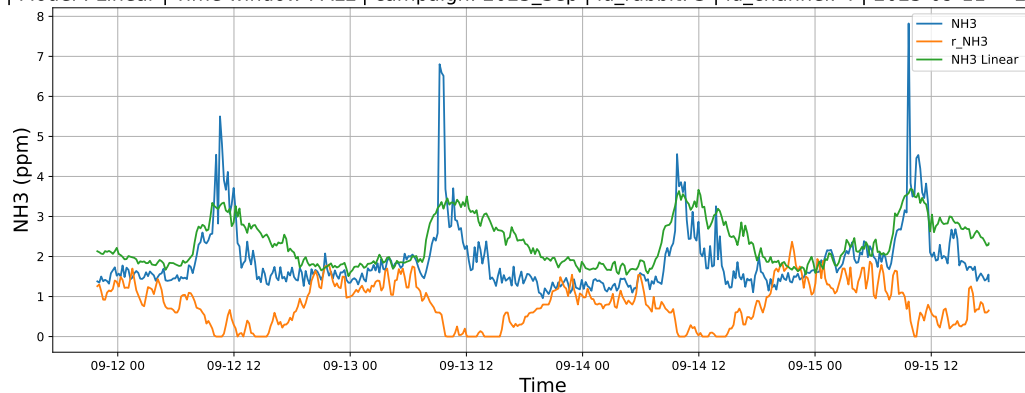


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

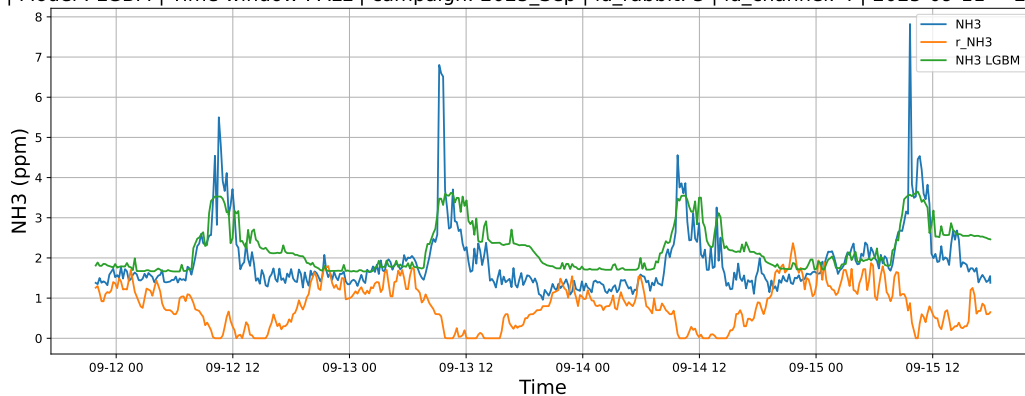


4.1.6.3 Time window : ALL

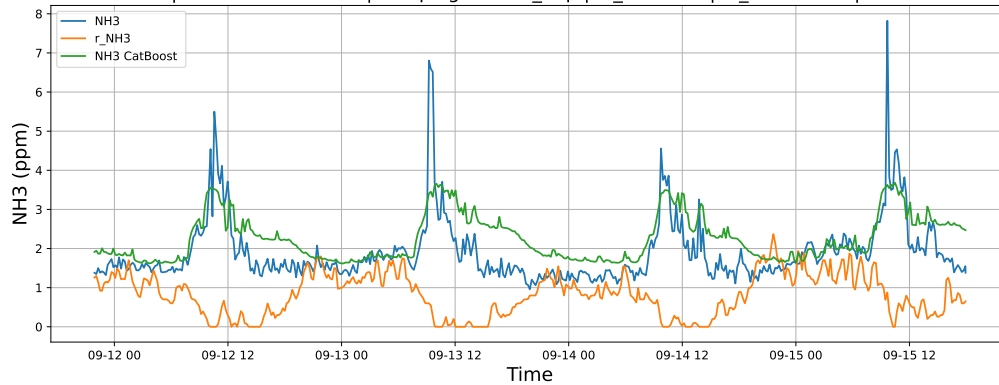
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



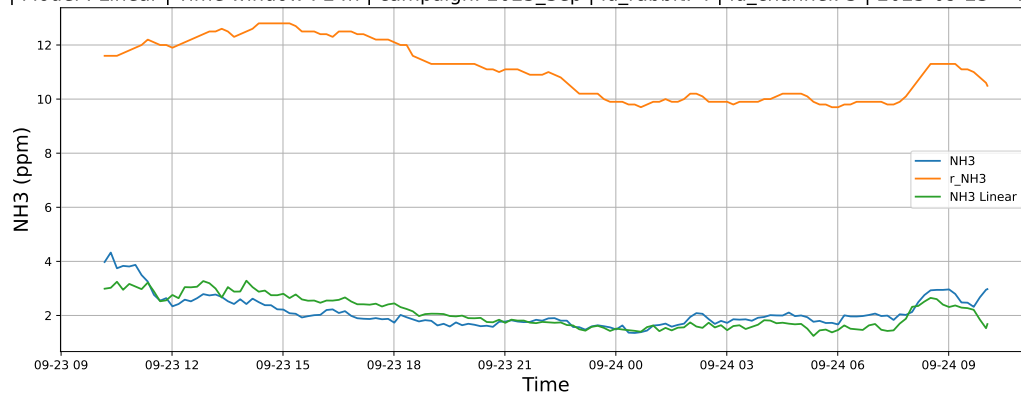
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



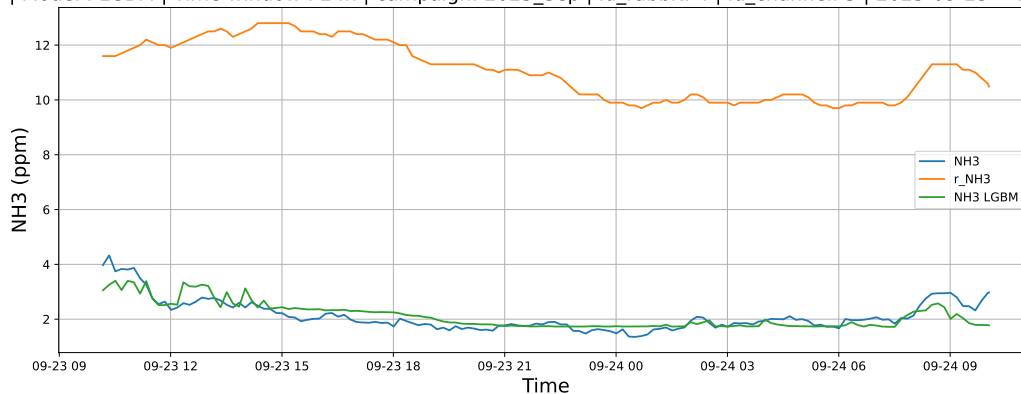
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

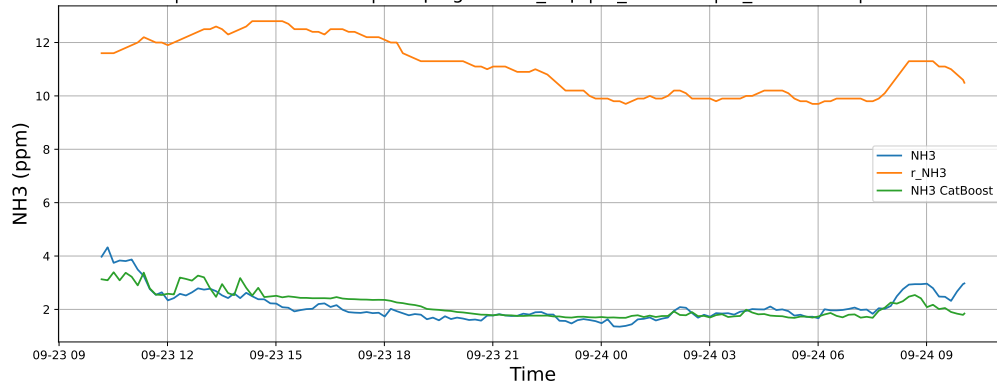
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

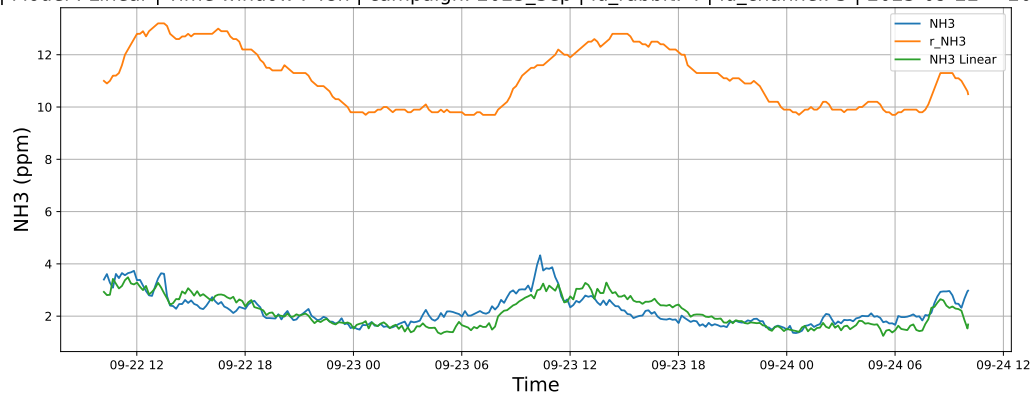


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

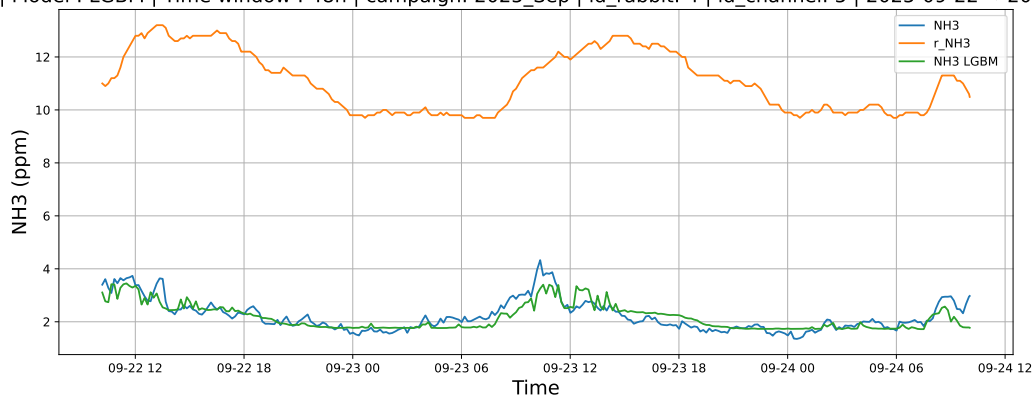


4.1.7.2 Time window : 48

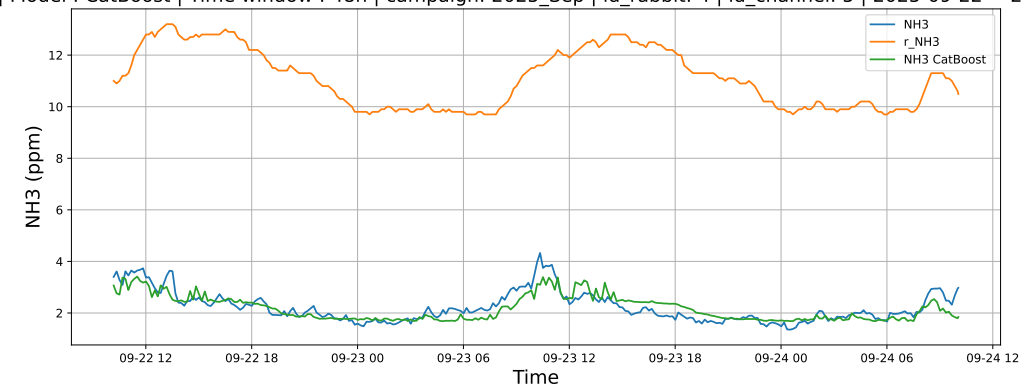
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

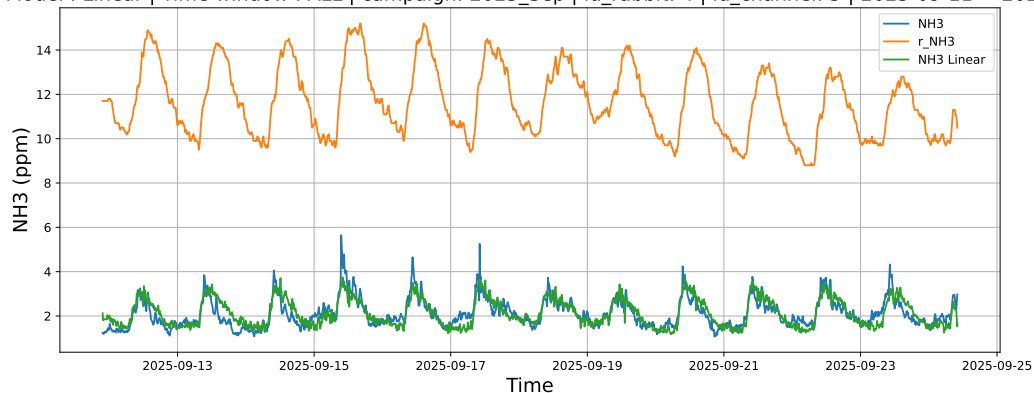


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

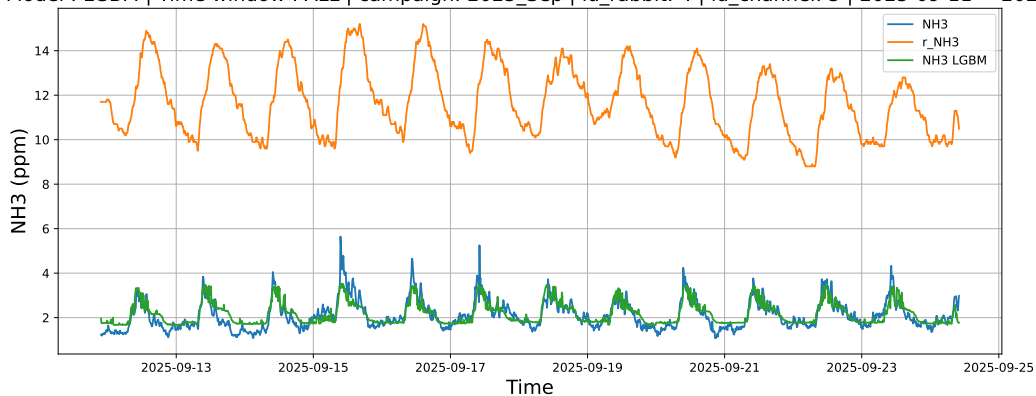


4.1.7.3 Time window : ALL

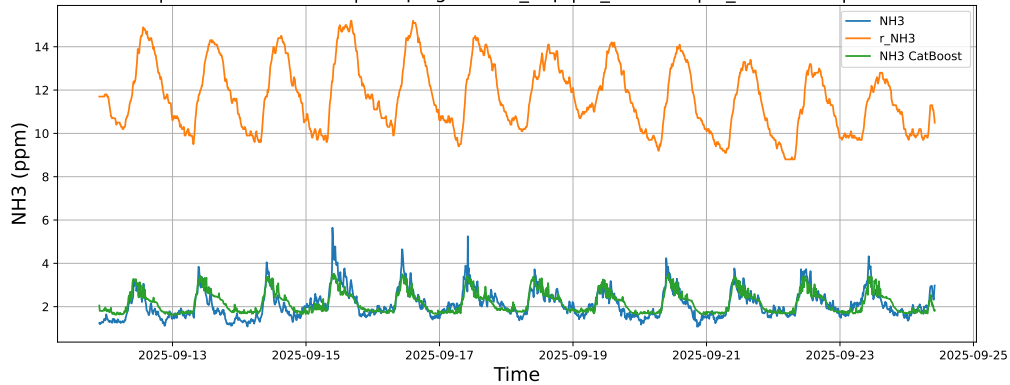
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



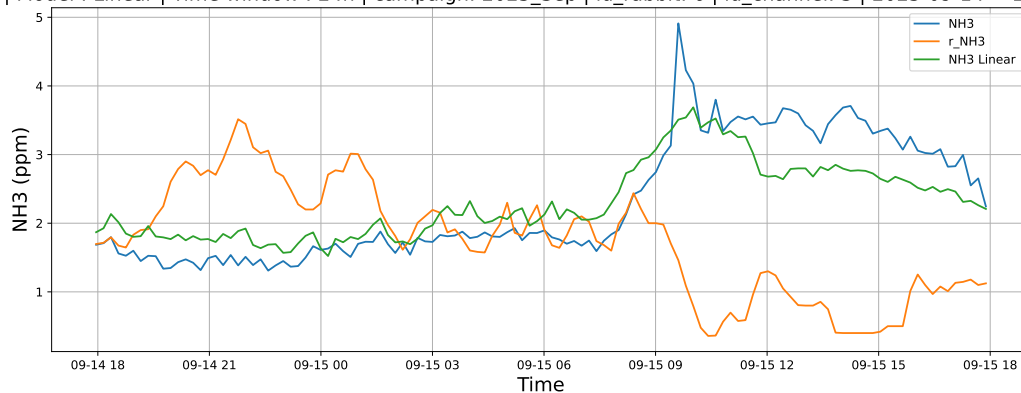
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



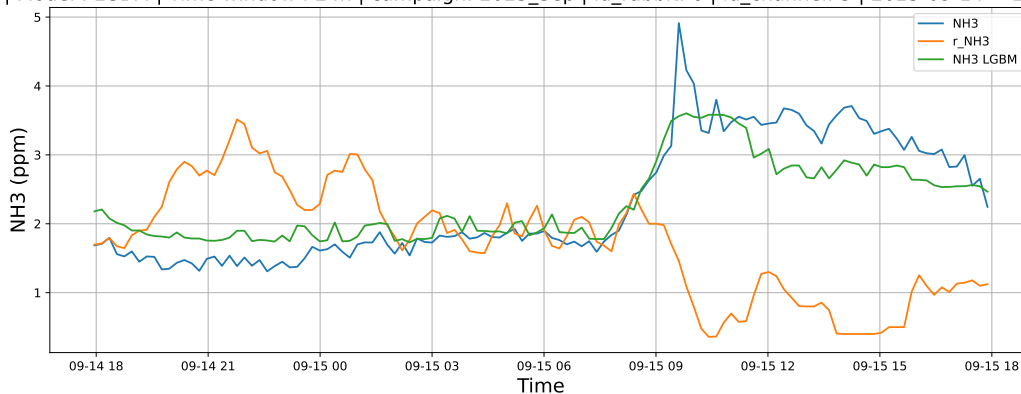
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

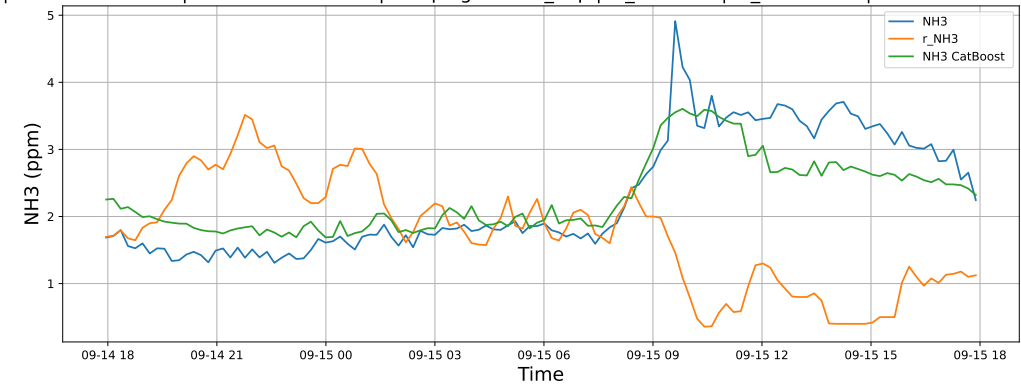
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

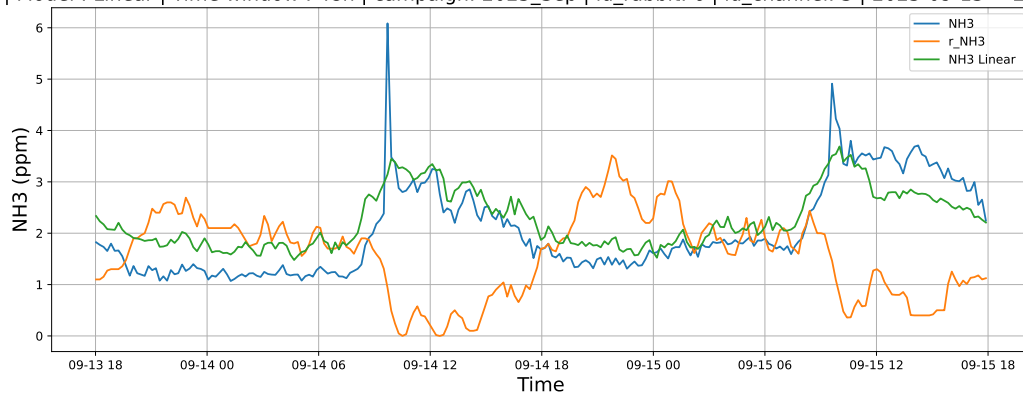


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

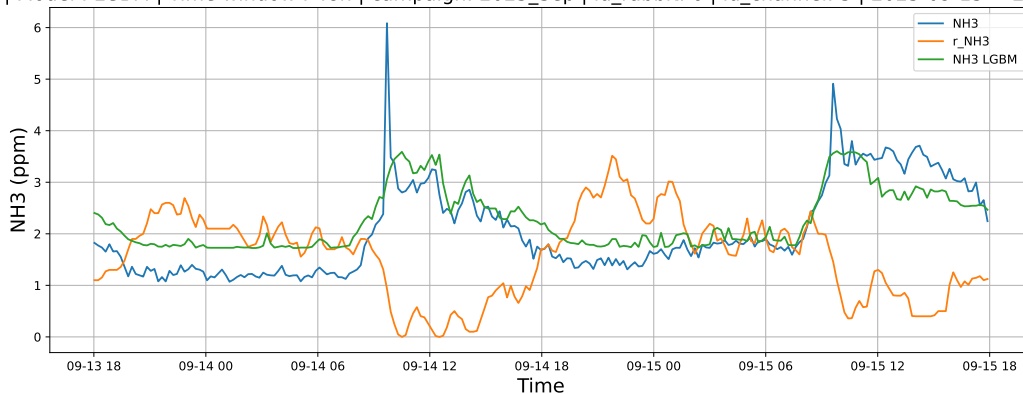


4.1.8.2 Time window : 48

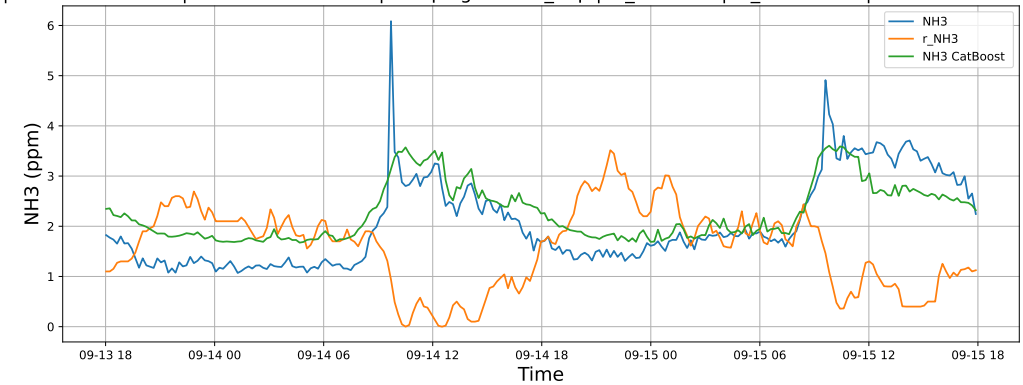
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

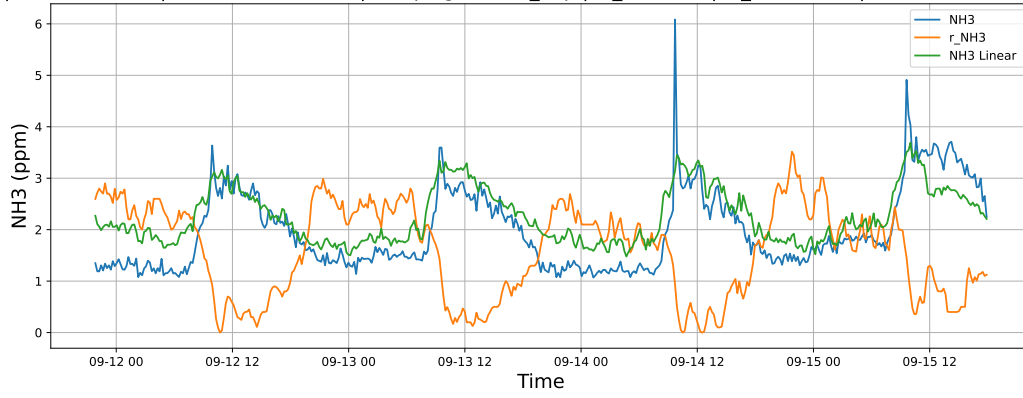


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

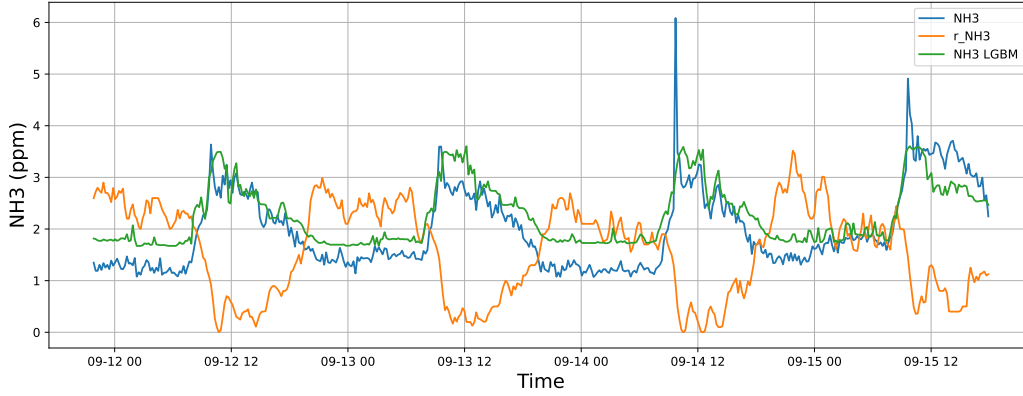


4.1.8.3 Time window : ALL

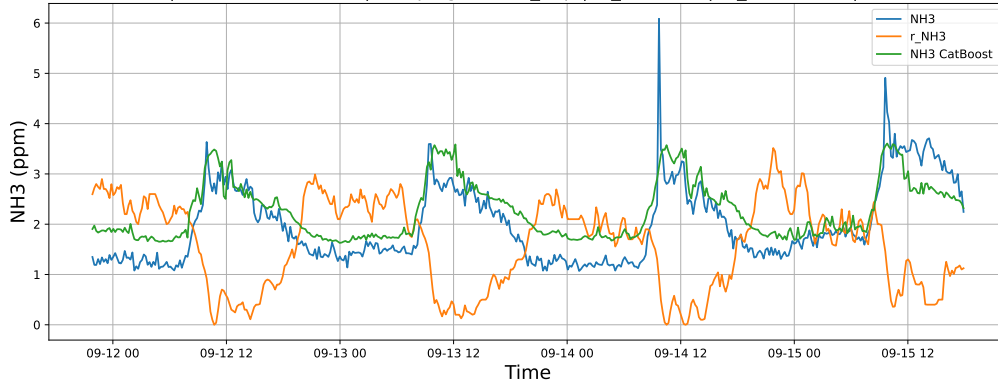
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

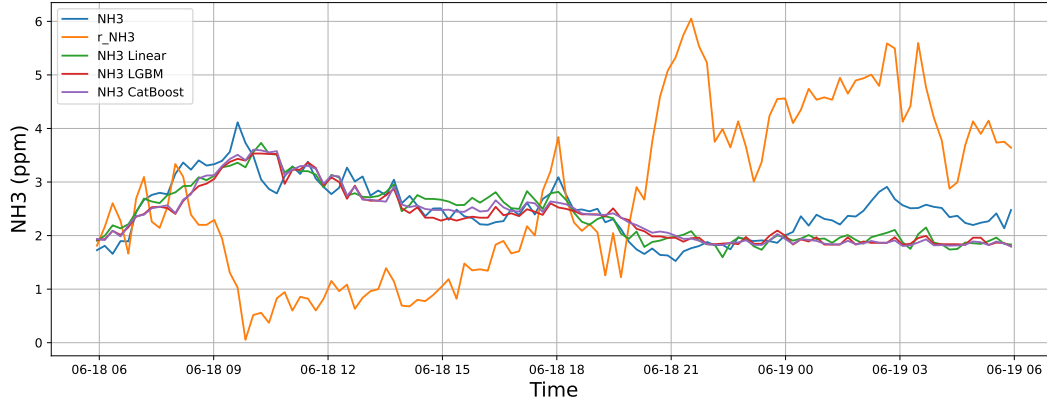


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

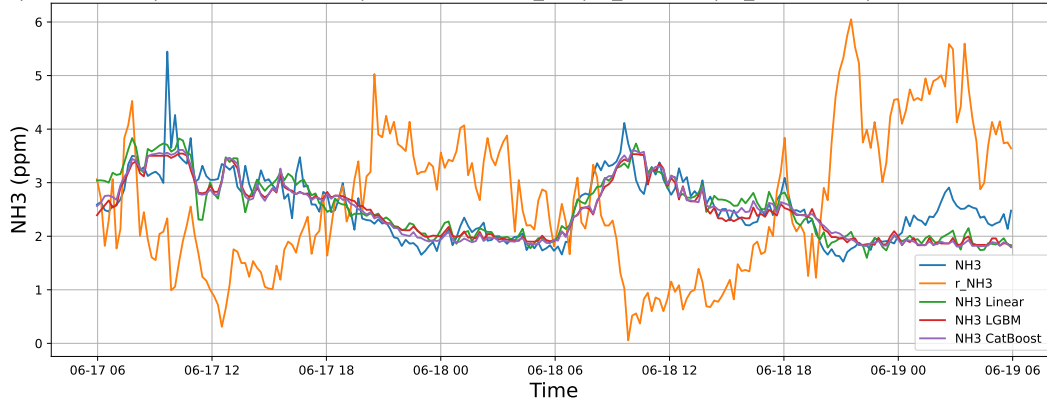
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



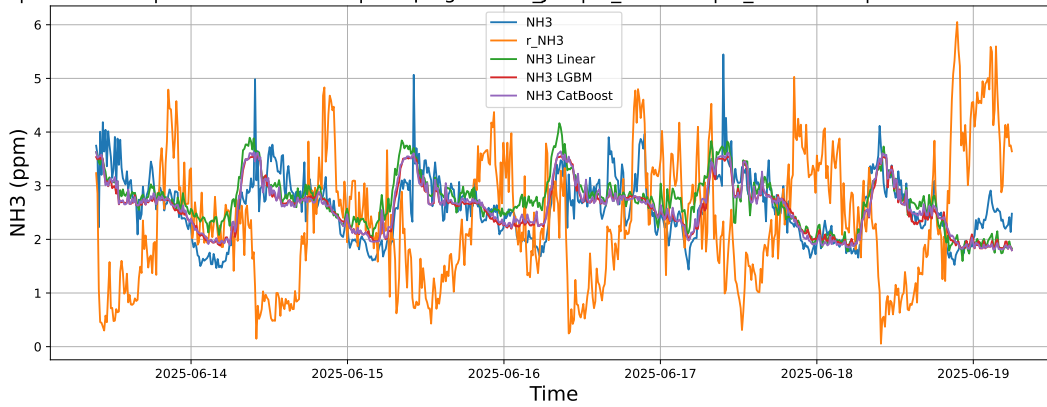
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

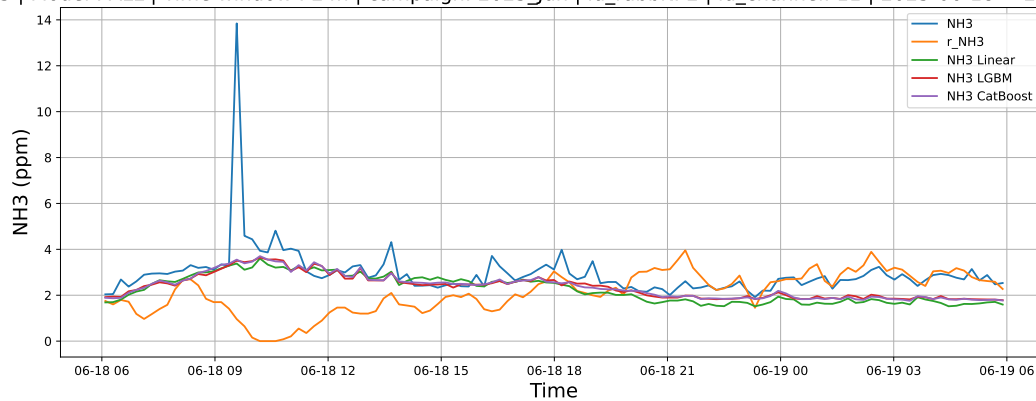
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

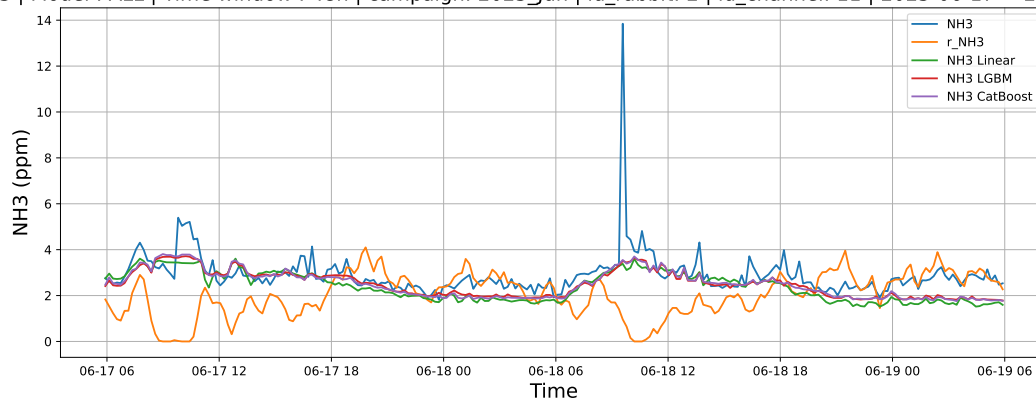
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



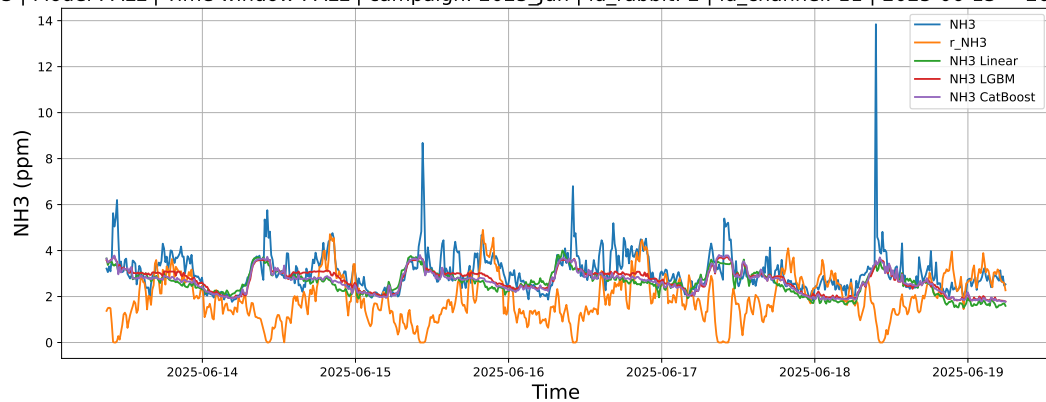
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

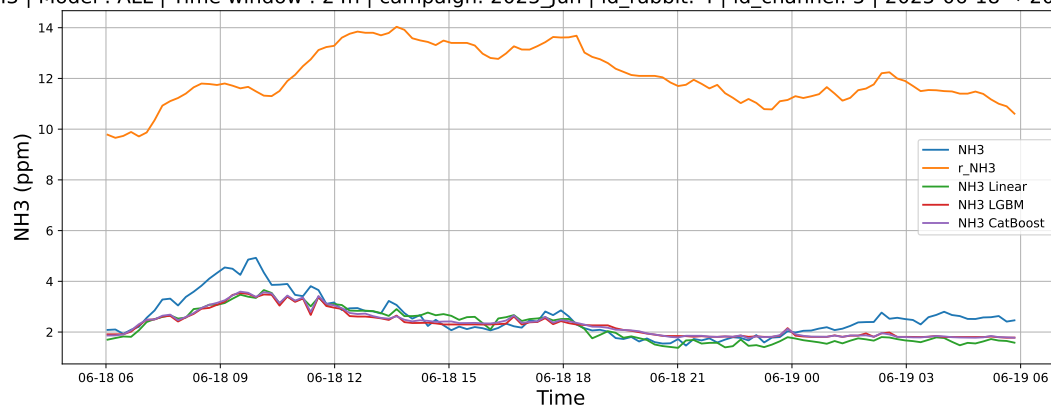
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

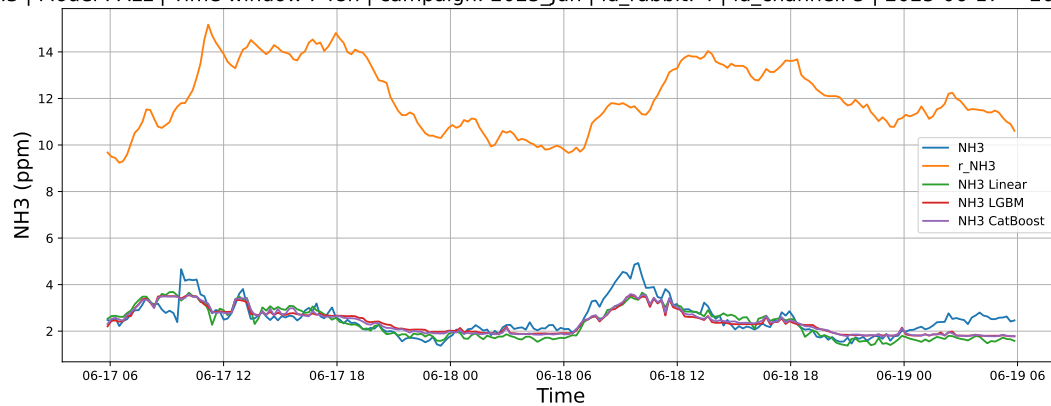
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



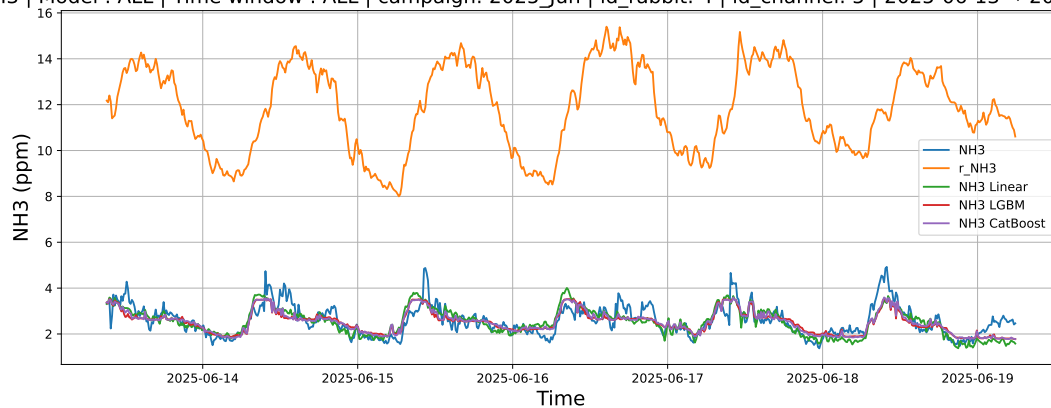
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

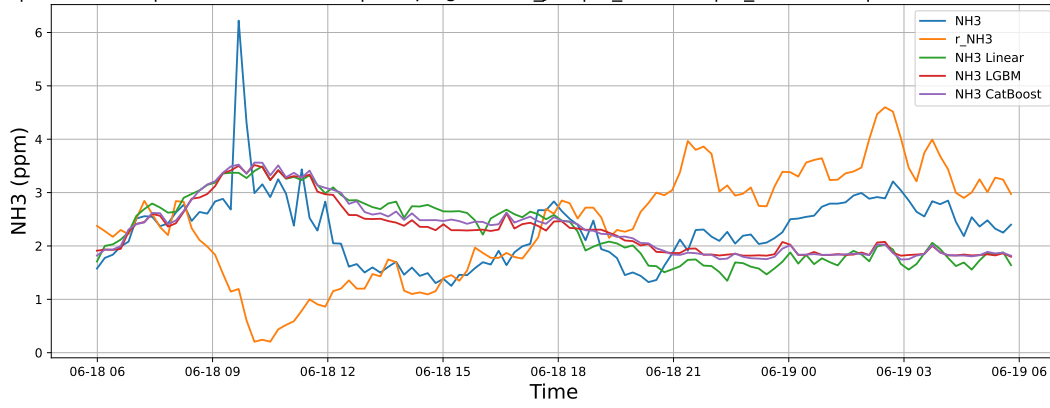
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

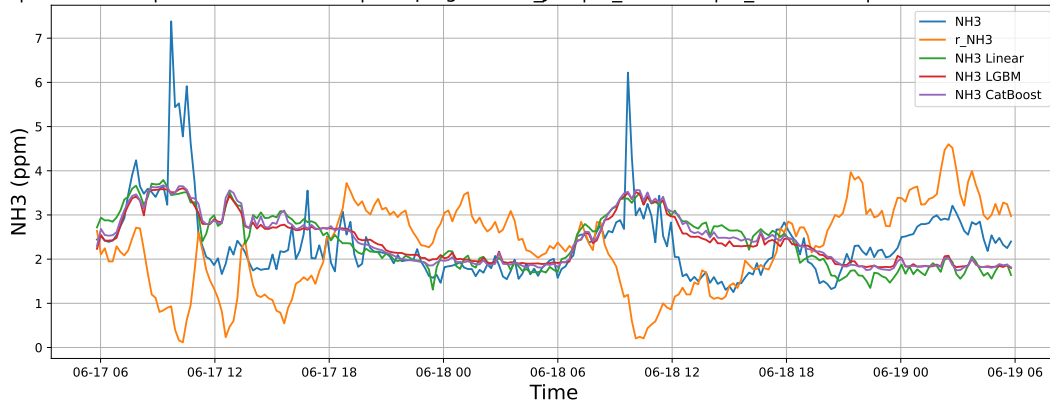
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



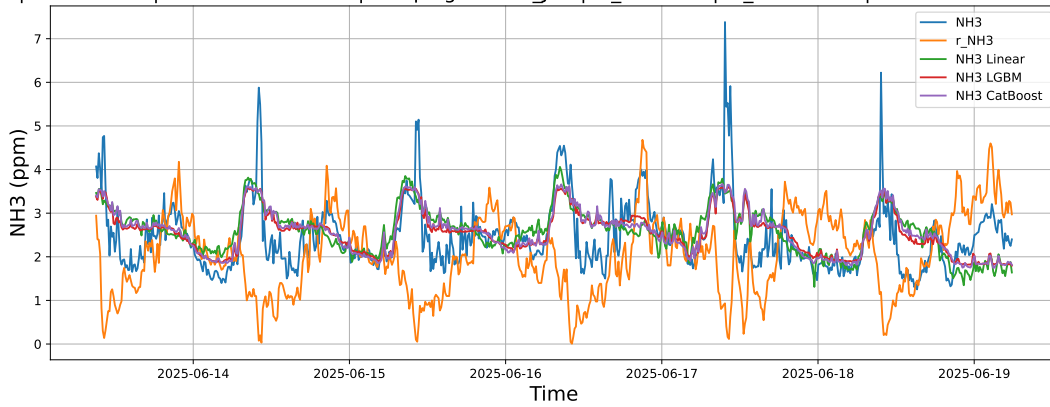
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

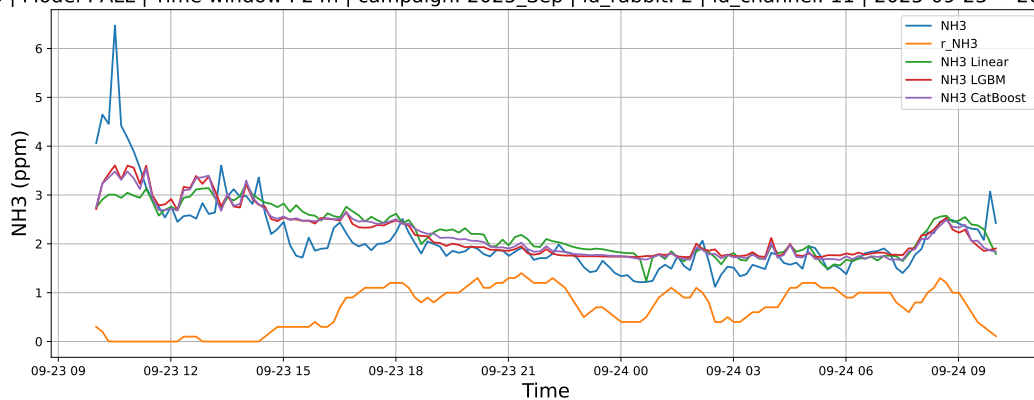
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

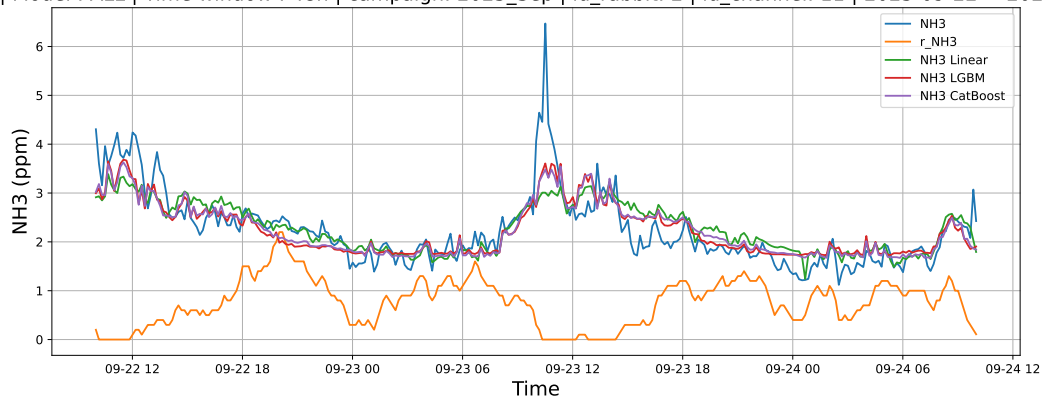
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



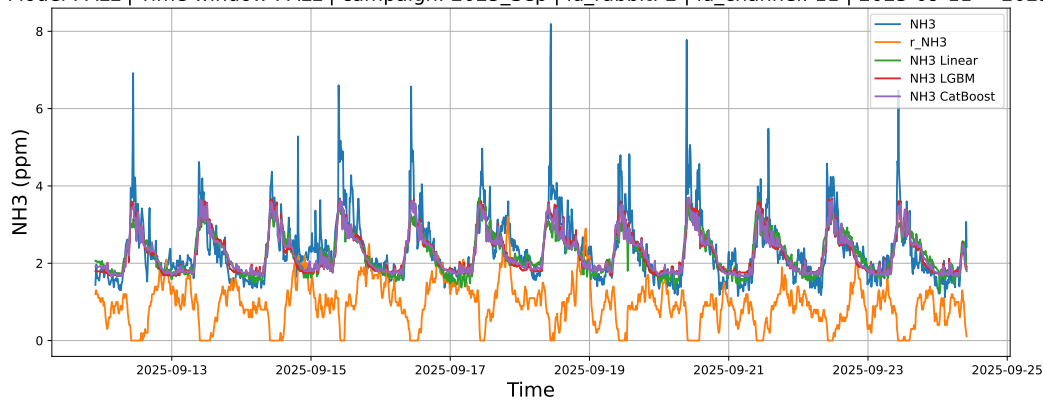
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

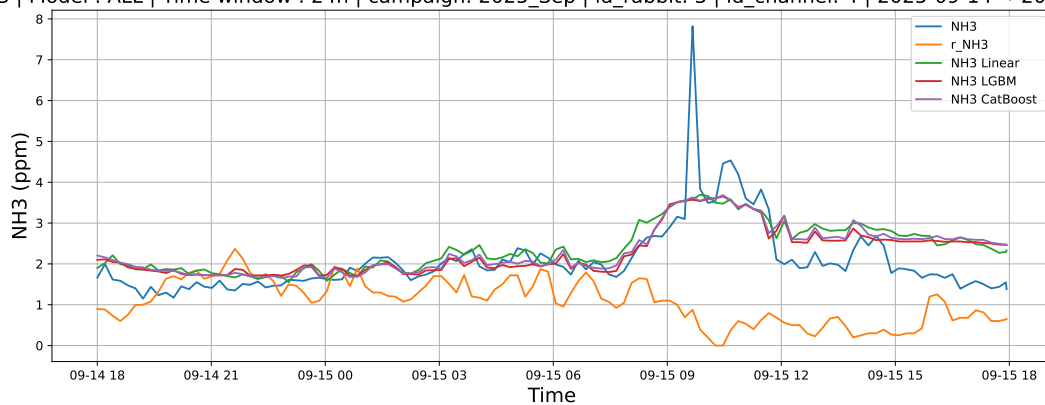
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

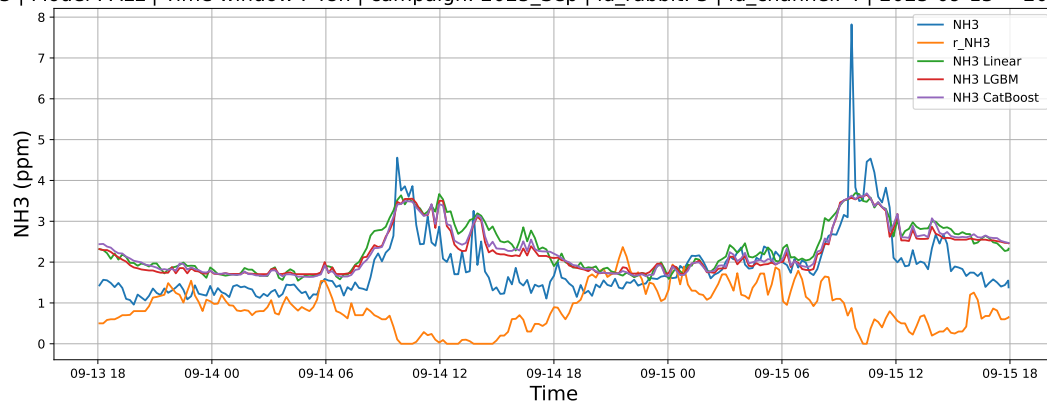
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



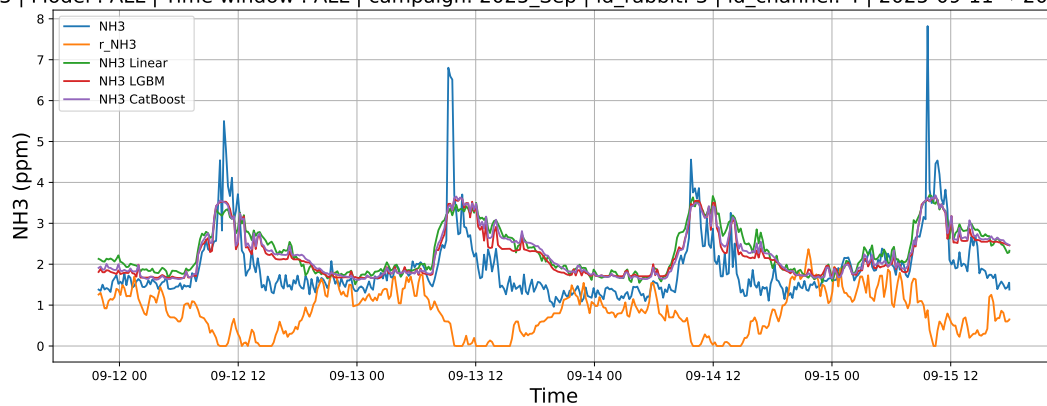
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

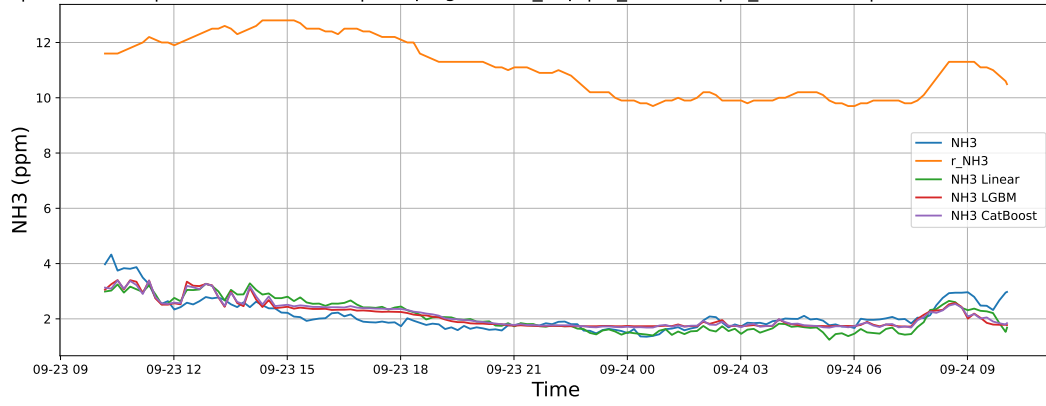
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

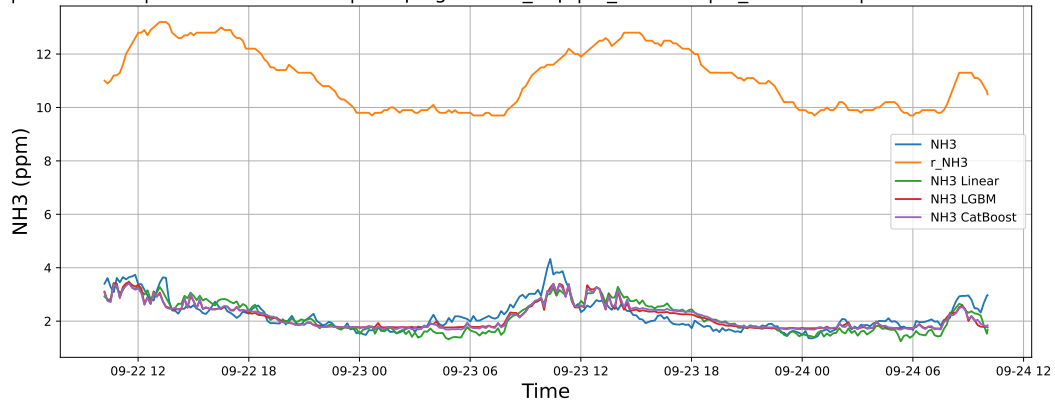
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



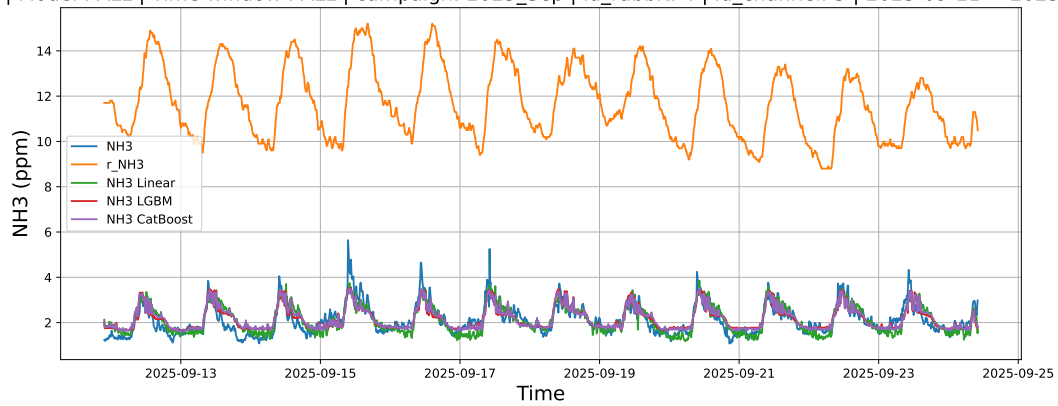
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

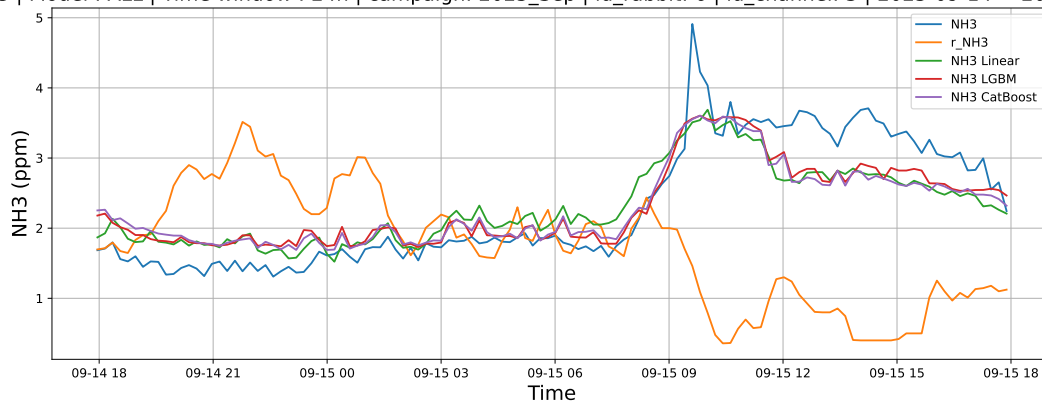
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

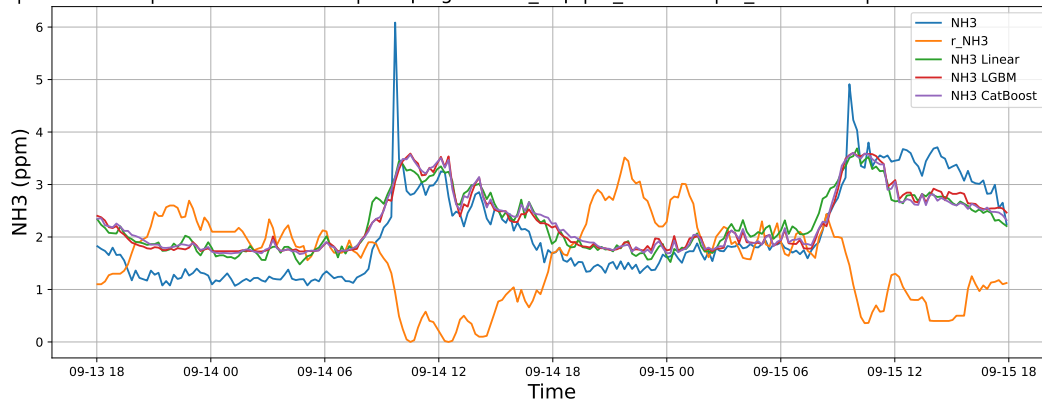
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

